Global Finance Journal 56 (2023) 100771

Contents lists available at ScienceDirect

Global Finance Journal

journal homepage: www.elsevier.com/locate/gfj

Noninterest income mix and bank performance: Generalized propensity score fractional dose–response function evidence☆

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ARTICLE INFO

Keywords:
Dose–response function
Generalized propensity score
Noninterest income
Revenue diversification

ABSTRACT

This paper investigates the impact of different levels of noninterest income on the risk-adjusted profitability and financial stability of banks in the Middle Eastern and North African banking industry from 1988 to 2020. We apply a program impact evaluation analysis, using the generalized propensity score-fractional dose–response function method, and assess the effect of each level (or dose) of noninterest income mix on bank performance (the dose–response function). Splitting the sample into Gulf Cooperation Council- and non-Gulf Cooperation Council-based banks reveals that while higher levels of revenue diversification are beneficial for the former, a focus on traditional banking activities is preferable for the latter to achieve the best performance. The findings could serve as an early warning flag for bank managers, policy-makers, and regulators in dealing with these strategic shifts.

1. Introduction

Financial innovation, deregulation and financial liberalization, policy reforms, technological progress, and the development of managerial skills have spurred banks worldwide to drastically change the nature of their business models. These profound changes, coupled with increasing competition among banking firms in loan and deposit markets, have created incentives for banks to shift away from traditional activities toward new market segments that have generated commissions (e.g., credit cards, investment advice, business consulting services), trading income (e.g., trading of government securities, financial derivatives, trading of private securities, equity securities) and other noninterest income (e.g., fiduciary activities). The increasing shift toward nontraditional activities was thought of to be profit-boosting (Baele, De Jonghe, & Vander Vennet, 2007; Chiorazzo, Milani, & Salvini, 2008) via economies of scope (i.e., selling new products such as financial consulting services to their existing customer base, mainly loan clients) and scale (i.e., the sharing of inputs, such as human capital, information, or technologies across multiple outputs, which is a major source of cost savings). This strand of empirical and theoretical literature also highlighted the benefits of widening the scope of bank revenue streams as seeking new market segments would increase the competitive nature of banking firms and mitigate their risks (Goddard, McKillop, & Wilson, 2008) since noninterest revenue was uncorrelated (or at least imperfectly correlated) with interest income (Sawada, 2013). Although arguments in favor of these strategic shifts dominated before the financial turmoil, the 2007–2009 Global Financial Crisis (GFC) reminded both researchers and practitioners about the utmost need to assess the benefits and drawbacks of this strategic

☆ The authors would like to thank the GFJ editor and the anonymous referees for their insightful comments, suggestions and careful reading of the manuscript. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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https://doi.org/10.1016/j.gfj.2022.100771
Received 7 February 2022; Received in revised form 4 August 2022; Accepted 6 August 2022
Available online 5 September 2022
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decision. The subprime crisis revealed a dark side of diversification, which was especially related to market-oriented (trading) activities and to the lack of regulation (De Jonghe, 2010; Brighi & Venturelli, 2014), which in turn would make the reassessment of this practice of urgent need.

Opponents of revenue diversification believe that the latter may generate agency problems between managers and shareholders due to the complexity of the cost structure and to the organizational form of the diversified banks (Laeven & Levine, 2007; Berger, Hasan, & Zhou, 2010; Guerry & Wallmeier, 2017). Shifting away from off-balance sheet activities is also due to the instability of noninterest income as trading-generating activities are market-oriented and are highly affected by market fluctuations (Baele et al., 2007; Calmès & Théoret, 2010). More than that, the high fixed costs associated with nontraditional activities increase the operating leverage of banks which in turn raises the volatility of the earnings of the banking firms. Furthermore, shifting towards nontraditional activities calls for higher levels of leverage compared with lending, as fees do not require raising additional regulatory capital (DeYoung & Roland, 2001).

The debate on the preference of universal versus specialized banking models has been broadly debated either for developed economies (e.g. Chiorazzo et al., 2008; Elsas, Hackethal, & Holzhäuser, 2010; Stiroh, 2004a, 2004b; Stiroh & Rumble, 2006) or for their developing/emerging peers (Berger et al., 2010; Lee, Yang, & Chang, 2014; Lee, Hsieh, & Yang, 2014; Meslier, Tacneng, & Tarazi, 2014; Ammar & Boughrara, 2019a; Ammar & Boughrara, 2019b). Yet, no consensus has been reached so far. The inconclusiveness of findings seems to be deeper than just disparities in banking business models or differences in the macroeconomic environment in which banks operate. Digging more into this topic seems to be of paramount importance not only for academics but also for bank managers, regulators, and policy-makers, giving them insights into how to deal with these strategic shifts.

Although the financial deregulation steps were undertaken in Middle Eastern and North African (MENA) countries, no profound changes have occurred in the business models of banks set in this region yet. MENA banking firms are generally focused (Abuzayed, Al-Fayoumi, & Molyneux, 2018; Ammar & Boughrara, 2019a), even those that operate in developed financial systems and where they are allowed to combine traditional banking activities and nonbanking financial business lines, such as the Gulf Cooperation Council (GCC) banking industry. The question of how universal banking has affected the performance of MENA banks has been debated in the literature (e.g. Abuzayed et al., 2018; Ammar & Boughrara, 2019a; Ammar & Boughrara, 2019b). Still, a thorough analysis is of a paramount importance for regulators and policy-makers who care about the soundness and stability of the MENA banking sector. The sustainability of the financial sector crucially relies on the stability of the banking sector, especially for bank-based systems such as the MENA industry.

We contribute to the extant literature in the following way. We employ a new estimation approach that allows us to bring explicit answers to the ongoing debate on the impact of revenue diversification on bank performance. To do so, we approach the data somewhat differently and focus on the heterogeneity of diversification impact on the risk-adjusted profitability and financial stability of the MENA banks that may arise from variations in the level (or dose) of noninterest income mix. So far, the plausible reason for the disparity of findings in previous empirical investigations is that the latter did not consider the effects of different levels of noninterest revenue on bank performance. All the papers that have been published so far have either considered aggregate and/or categories of noninterest income (e.g. Stiroh & Rumble, 2006; Chiorazzo et al., 2008; Elsas et al., 2010; Ammar & Boughrara, 2019a), but no one has considered the impact of noninterest income doses.

Hence, in contrast to the traditional econometric approaches carried out in the empirical literature, we distinguish ourselves by providing a new approach that gauges the performance of MENA banks at each level of income mix. As far as we know, this is the first study that uses the Generalized Propensity Score (*GPS*) approach in the bank performance–diversification literature. This methodology is an extension of the mainstream Propensity Score Matching (*PSM*) for binary treatments of Rosenbaum and Rubin (1983). Its advantage over the latter is that it better suits the real world conditions, since, in social science, the treatments generally take a continuum of values and are not binary.

Compared with the mainstream econometric approaches, *GPS* is increasingly employed to mitigate the confounding bias that occurs in nonexperimental studies.[1] Said differently, adjusting for the *GPS* provides an alternative way to remove all the bias associated with the differences in prediversification decision covariates between MENA banks belonging to the treatment (diversified) and the control (focused) groups. The assessment of the heterogeneity of the diversification impact amounts to estimating a Dose–Response Function (*DRF*). Thus, we estimate the response (i.e., MENA bank performance) that corresponds to particular values of the continuous treatment doses (i.e., levels of noninterest income mix). Also, apart from a clear extension to the scope of the extant literature, this empirical investigation focuses on the MENA banking industry that needs a particular attention and for which an in-depth innovative analysis is of the utmost importance. The method proposed in this paper offers an opportunity for good quality research on impact evaluation in general and promises to provide new insights into the bank performance–revenue diversification nexus in particular when the exposure to the treatment (noninterest income mix) is continuous.

As no consensus has yet been reached, our study aims to bring an explanation to the revenue diversification bank performance nexus. To do so, we try to answer the following question: What is the impact of different levels (doses) of noninterest income mix on the performance of MENA banks ?.

Our evidence suggests that revenue diversification creates value for GCC-based banks that attain their highest performance via the maximum noninterest income mix. For non-GCC banking firms, shifting away from noninterest income generating business lines and

[1] The empirical methodology we implemented in this paper mainly relies on Guido Imbens' research (e.g., Imbens, 2000; Hirano & Imbens, 2005; Abadie & Imbens, 2006; Imbens & Wooldridge, 2009; Imbens & Rubin, 2015). It is worth stressing that Guido Imbens has been awarded the 2021 prize Nobel in Economics Sciences for his methodological contributions to the analysis of causal relationships.

relying on mainstream traditional activities is still of a paramount importance as the best risk-adjusted profitability and the highest financial stability are reached at the lowest levels of noninterest revenue mix (and lowest noninterest income share).

The remainder of this paper is organized as follows. The next section provides a review of the relevant empirical literature on the bank performance–revenue diversification nexus. Section 3 describes the data used. In Section 4, we outline and detail the methodology employed. Then, we discuss the empirical findings (along with the robustness checks). Finally, Section 6 concludes and provides insights for bank managers, regulators, and policy-makers about how to deal with revenue diversification.

2. Diversification strategy: Merits and demerits

Three streams of view prevail in the extant literature. While the first strand highlights the benefits of shifting toward off-balancing sheet activities, the second rather foregrounds the drawbacks of doing so. The last strand gives evidence of a mixed impact of revenue diversification on bank profitability and risk.

The concern of restoring profitability and boosting the stability of the banking sector have led to several calls for an expansion toward nontraditional business lines. Proponents of the strategic decision argue that the diversification of revenue mix improves the risk-return trade-off by stabilizing bank profit streams (Chiorazzo et al., 2008) and mitigating the overall riskiness (Sawada, 2013), given that noninterest income-generating activities are uncorrelated (or at least imperfectly correlated) with interest-based business lines. In the same vein, a developed cross-country fixed-effect panel model evidence provided by Elsas et al. (2010) revealed the existence of a diversification premium impacting indirectly and positively bank market valuation, via profitability. A later study argued that highly-leveraged large US bank holding companies may benefit from financial consolidation in the form of lower specific risk and greater profitability (Demsetz & Strahan, 1997). Considering 52 Italian bank holding companies, Brighi and Venturelli (2014) proved the existence of a positive impact on adjusted and unadjusted bank profitability while no significant effect on risk was reported.

There has been a growing body of literature that focused on emerging/developing economies, stressing the presence of diversification benefits. While investigating the impact of the ownership structure, Pennathur, Subrahmanyam, and Vishwasrao (2012), via their pooled model and random-effect panel data regressions, found that expanding bank business lines had the merit of mitigating risks for foreign banking firms. As illustrated by Lin, Chung, Hsieh, and Wu (2012), the smoothing effects on the performance of Asian banks were caused by the role played by diversification in mitigating the sensitivity of the net interest margin to risk fluctuations. This finding was obtained via a switching function in which the noninterest margin equation belonged either to the category of "high degree of diversification" or "low income diversification." Diversification merits took the form of higher profit gains, higher market shares (Nguyen, Skully, & Perera, 2012a), economies of scale benefits (Nguyen, Skully, & Perera, 2012b), higher capital savings and lower insolvency risk (Shim, 2013). Nguyen et al. (2012a) reported that banks benefited from these strategic shifts via higher profit gains and when their market shares in the deposit and loan markets were low, especially after a period of financial meltdown. Nguyen et al. (2012b) fixed effects model revealed that large South Asian banking firms had more incentives to engage in nontraditional activities when they operated in economies with a high financial development. Using System Generalized Method of Moments (GMM) estimation, Lee et al. (2014) highlighted that shifting towards off-balance sheet activities had the potential to boost Asian-pacific banks' profitability and stability, especially for bank-based financial systems. Using fixed-effect and dynamic panel estimation models, Meslier et al. (2014) noticed that universal banks, especially foreign ones, enjoyed greater profitability and lower risk. The authors also reported that market-based business lines that generated trading income had a smoothing impact as they were the least correlated with interest revenue. Guerry and Wallmeier (2017) documented a decline in the diversification discount in the valuation of US banks before the 2007–2009 financial crisis that vanished afterward.

Opponents of these strategic shifts tend to support the existence of diversification drawbacks. The aforementioned arguments were contradicted by Kwast (1989) who found limited potential of risk reduction for trading securities. This finding was supported by Kwan (1998) who underlined that section-20 subsidiaries were riskier than US banks commercial affiliates. Berger and Ofek (1995) provided an in-depth analysis of the role of agency problems in aggravating the situation as diversification would lead to inefficient cross-subsidization across segments, making poor segments to take advantage of their better performing counterparts. In their study, DeYoung and Roland (2001) provided strong evidence that fee-generating activities worsened the risk-return trade-off of US commercial banks. In fact, an increase in fee share would raise bank operating leverage and financial leverage, leading to an increase in earnings volatility. Stiroh (2004a) found no obvious benefits of revenue stream diversification. Rather, risk-adjusted performance would diminish as banks shifted toward trading-generating business lines (Stiroh, 2004b). This finding was corroborated by Stiroh (2006a, 2006b) who illustrated that increased reliance on nontraditional activities increased risk. In the same line of arguments, Stiroh and Rumble (2006), using cross-sectional and panel data, reported that the earnings gains owned via diversification were offset by volatility increases driven by the increased exposure of US financial holding companies to noninterest income. The authors highlighted this as the main drawback of revenue diversification. Consistent with US studies, Mercieca, Schaeck, and Wolfe (2007) did not find any benefit for small European credit institutions that would diversify their revenue streams. The diversification discount was also illustrated in the cross-country study of Laeven and Levine (2007) for financial conglomerates. Employing the instrumental variables approach as well as the Heckman (1979) selection model to counter the endogeneity of the diversification proxy, the authors tended to support the argument stipulating that economies of scope were offset by diversification drawbacks. The expansion toward new market segments was strongly related to risk increases (Lepetit, Nys, Rous, & Tarazi, 2008b). More than that, a specific shift toward commission and fee-generating business lines raised European banks' propensity to incur a default risk (Lepetit, Nys, Rous, & Tarazi, 2008a). This finding was supported by De Jonghe (2010) who, using both accounting and market data, demonstrated that financial consolidation would mitigate the European banking system stability via higher bank tail betas. Focusing on the Australian banking industry, Williams (2016) proved the presence of revenue volatility increases as well as systemic risk in response to increased shift

toward off-balance sheet activities.

The last strand of the empirical literature on bank performance-revenue diversification nexus provides mixed effects. Acharya, Hasan, and Saunders (2002) has provided evidence of an improvement in the risk-return trade-off for Italian commercial banks with a moderate risk exposure. Using Ordinary Least Squares (OLS) regressions, Baele et al. (2007) considered the link between nontraditional activities and risk premium required to evaluate the risk inherent to these new business lines. According to the authors, although shifting toward off-balance sheet activities had the potential to improve European banks' profitability, an aggressive diversification strategy increased the systemic risk for large banks. In fact, Baele et al. (2007), based on cross-sectional data, revealed that noninterest income-generating activities were more market-oriented compared to traditional income sources, and so were more subject to market and macroeconomic shocks. As noted by Goddard et al. (2008), who employed cross-sectional and instrumental variables (VI) regressions, entering new market segments that generated noninterest income boosted the financial stability of highly diversified US credit unions. Yet, noninterest income was still more volatile than interest revenue. Diseconomies of diversification diminished in importance when the foreign ownership stake was taken into account (Berger et al., 2010). Demirgüç-Kunt and Huizinga (2010) examined how banks' income mix influenced the bank return rate and the z-score (or distance to default).

The cross-country analysis revealed that the risk-return trade-off, in terms of bank return rate and risk, went up with fee income share. The authors concluded that, for low levels of noninterest income, there could be some risk diversification benefits. However, for higher exposures to nontraditional activities, the bank risk status exacerbated. Focusing on the Canadian banking sector and using an ARCH-M estimation approach, Calmès and Théoret (2010) came to the conclusion that the negative impact of off-balance sheet activities on bank returns was mitigated in the 1997–2007 period. The impact of size was considered by Hidayat, Kakinaka, and Miyamoto (2012) who stressed that large Indonesian banks that aggressively diversified their revenue streams incurred higher risks. The authors found that increases in the commission and fee share adversely impacted the risk level of small banks. According to the multi-period logit model of DeYoung and Torna (2013), diversification did not build up the insolvency risk per se. Rather, a rise in risk was linked to the nature of income in consideration. The authors also demonstrated that diversification raised the probability of bank failure during the GFC. According to Lee et al. (2014) dynamic panel GMM model, the impact depended on the bank type and the country's income level. De Jonghe, Diepstraten, and Schepens (2015), using fixed effects and IV approaches, showed that the positive impact of revenue diversification on bank risk diminished with size. The authors stressed that the benefits of these strategic shifts were closely linked to the country' specific institutional characteristics for large banks. Abedifar, Molyneux, and Tarazi (2018) modeled the relationship between noninterest income and credit risk by performing dynamic-panel and fixed-effect methods. The authors highlighted that the joint production had an adverse impact on US small banks and that fiduciary income and credit risk had a negative relationship in the postcrisis period. A recent empirical investigation of Kim, Batten, and Ryu (2020) has revealed that a moderate shift toward nontraditional activities would increase bank stability only to a certain threshold, using a rich set of estimation approaches.

Lately, the notion of universal banking and financial consolidation has been extensively debated for the MENA banking industry in the empirical literature. To begin with, Mnasri and Abaoub (2010) found that while revenue diversification had a smoothing impact on banks' profitability, it exposed them to greater systemic risk. Using a quantile regression, Rajhi and Hassairi (2013) concluded that these strategic shifts boosted financial performance but made IBs less stable. In contrast to the three-stage least squares approach of Lassoued and Sassi (2017) which stressed a risk reduction benefit of diversification, the random-effect panel data model of Ashraf, Ramady, and Albinali (2016) showed that the increasing shifts toward off-balance sheet activities raised the risk-taking attitude of banks with a concentrated ownership structure. Focusing on the GCC banking system and using the system GMM estimation approach, Abuzayed et al. (2018) argued that diversification did not boost the GCC-based bank financial stability. Running the same estimation approach, Ammar and Boughrara (2019a) demonstrated that diversification, when taken as a whole, boosted MENA banks' profitability, especially via trading-generating business lines. However, the authors documented that commissions, fees, and other noninterest income worsened the benefit-cost trade-off of diversification. Likewise, Ammar and Boughrara (2019b) showed that profitable GCC banks with a low credit risk had more incentives to diversify their revenue streams, proving the positive nexus between bank performance and these strategic shifts. Finally, AlKhouri and Arouri (2019) focused on the impact of diversification on GCC banks' stability. The authors evidenced that the impact differed according to bank type. While income diversification had an adverse effect on the stability of both conventional and Islamic banks, the impact was found to be particularly pronounced for conventional GCC-based banking firms.

The overwhelming majority of the existing literature that studied the impact of combining banking and nonbanking financial activities under one umbrella institution used classical econometric approaches. Hence, we differentiate ourselves by proposing a new method that considers the impact of different levels (or doses) of diversification. We believe that although the disparity of results is partly due to differences in the bank business models and the macroeconomic environment, the perception of the diversification proxy (i.e., considering aggregate, categories or doses of noninterest income) is the major factor that drives this inconclusiveness. We aim to solve the puzzle of the revenue diversification-bank performance nexus and bring more clarity to the relationship of interest.

3. Data description and sources

3.1. Preliminary sample processing

Bank-level data stem from Bankscope, a database maintained by International Bank Credit Analysis Ltd. (IBCA) and from the Brussels-based Bureau van Dijk. The sample we constructed consists of annual bank balance sheet data covering the 1988–2020 period. The countries for which data are drawn are: United Arab Emirates, Egypt, Bahrain, Lebanon, Morocco, Oman, Algeria, Tunisia, Jordan, Kuwait, Qatar and Kingdom of Saudi Arabia. Some preliminary processing was performed before embarking on econometric treat-

ment. Statistical individuals (i.e., banks) for which data were missing for the required key variables were dropped. To mitigate the potential influence of outliers, the initial sample was winsorized at the conventional 5th and 95th percentiles. Likewise, to account for the potential reporting errors, all bank-observations with either negative assets or negative loans were suppressed. In the same vein, to make sure to perform the required tests (i.e., the second-order serial correlation test), all the statistical individuals with less than four consecutive years were taken off from the sample. Similarly, statistical individuals with significant number of missing values for more than two relevant balance sheet items were also dropped from the sample. After carrying out these filters, we ended up with a highly unbalanced panel comprising 370 banks covering the period ranging from 1988 to 2020.

3.2. Data description

The outcome variable used in this study is bank performance. It is proxied by three indicators, namely RAROAA, RAROAE and the z-score. RAROAA is the risk-adjusted return on average assets. It is measured by the return on average assets to the standard deviation of return assets. RAROAE is the risk-adjusted return on average equity. It is measured by the return on average equity to the standard deviation of return equity. We opt for adjusted measures for risk to account for the total volatility of profits. Z-score is a measure of the insolvency risk.[2] It is proxied by the average return on assets and equity to assets to the standard deviation of return assets. The program take-up is apprehended by using a continuous treatment indicator (i.e., the diversification indicator), which takes fractional values in the interval]0,1[whose intensity (or dose) signals whether the bank diversified its revenue streams (for doses that tend to the upper limit) or rather focused on its main traditional intermediary activities (in the opposite case). It stems from the Herfindahl–Hirschmann concentration index (div_non), which measures the diversification within the main components of noninterest income mainly, fees and commissions, trading income and other noninterest revenue (see Appendix A). Low (near to 0) or high values (near to 1) show that banking firms are not diversifying enough through noninterest income-generating products.

As for the covariates, we enumerate the confounding factors that are associated with bank performance and to the propensity to shift toward nontraditional activities. For this purpose, a set of bank-macroeconomic and industry-specific factors are included. To begin with, we include variables related to bank characteristics using accounting measures. Accounting data allows us to benefit from using a larger sample and to be comparable with the empirical works on the diversification-performance relationship. Bank size is measured by the logarithm of bank total assets. Large banks can reap economies of scale and scope benefits from diversification (Chiorazzo et al., 2008; Nguyen et al., 2012b). They can implement new strategies by developing new products and services that generate noninterest income since they have the best investments in technology and possess the best expertise in using and developing up-to-date risk management tools (Nguyen et al., 2012a; Meslier et al., 2014). A different point of view is highlighted by Chiorazzo et al. (2008). Small-sized banks might benefit from these strategic shifts as they are more flexible and could adjust their strategies rapidly to the changing economic conditions. Hence, we are uncertain about the impact of size on the diversification indicator. Capital ratio is proxied by the book value of equity to total assets. On the one hand, a high degree of capitalization signals a high risk aversion (Meslier et al., 2014). Risk-averse managers think twice before engaging in new investments. Hence, these banks that forgo investment opportunities might be less risky but do not generate high returns. On the other hand, large capital ratios reduce the probability for a banking firm's of incurring a financial distress and shelter it from unexpected market shocks (Baele et al., 2007; Shim, 2013). Hence, the impact of capital-to-assets ratio on bank performance is not straightforward. We also include liquidity ratio, which stands for the ratio of liquid assets to total assets. One the one hand, a higher liquidity ratio mirrors the bank's ability to meet its short-term financial obligations as they come due. The more liquid the bank is, the more it is able to maintain its financial sustainability and reduce the risk of occurrence of liquidity risk. On the other hand, banking firms with liquidity problems are generally subject to customer runs for being labeled as uncreditworthy. To preserve and widen their customer base, banks need to offer new lines of products and services to lessen their exposure to (or diversify away) the liquidity risk. Hence, the impact is unknown à priori.

We also account for the bank management variables. Banks that have the best skilled managers are eager to improve the quality of products and services by shifting toward nontraditional activities and undertaking the most efficient and profitable investment decisions. In the same vein, the degree of management efficiency and expertise considerably and positively influences the implementation of this strategic shift as it shelters banks from unexpected shocks (Baele et al., 2007; Lin et al., 2012). At the outset, we incorporate deposits ratio as approximated by the ratio of deposits to total assets. Banks that possess higher shares of deposits have a stable funding source to finance their activities and to develop new products and services. Greater deposits-to-assets ratio has a stabilizing effect on their income (Lee et al., 2014) and increases the bank's capacity to generate additional revenues. Hence, a positive impact is expected on bank performance. Likewise, we account for bank growth that we calculate as the annual growth rate of real bank assets (to measure assets growth). This variable mirrors the managers' risk attitude. It reflects the impact of rapid growth on bank strategic choices (Lee et al., 2014). We expect that greater assets growth leads to a more financial instability and to a lower bank performance. We control for net interest margin ratio, which we measure using interest income relative to earning assets. It reflects the profitability of traditional interest-generating business lines. An efficient lending and investment strategies that generate interest

[2] Although z-score is generally known as a measure of insolvency risk, it has been used in the empirical literature as a measure of financial stability (see, for instance, Boyd & Runkle, 2007; Beck, Demirgüç-Kunt, & Levine, 2007; Demirgüç-Kunt, Detragiache, & Tressel, 2008; Laeven & Levine, 2009). The appeal of such measure is that it requires only accounting information for their computation and, as such, can be applied to both listed and unlisted financial institutions. In addition to that, it combines accounting measures of profitability, leverage and volatility in a single measure. Hence, it fits the purpose of our paper (Chiaramonte, Croci, & Poli, 2015). It is worth reminding that higher (resp. lower) z-score may be interpreted as better (resp. lower) financial stability.

income enable banks to earn higher profit margins (Ammar & Boughrara, 2019a). However, unprofitable banking firms might expand toward new market segments that generate noninterest income to compensate their losses from interest-generating activities (Ammar & Boughrara, 2019b). In the same vein, we consider the credit risk by controlling for loan loss provision ratio (loan loss provision to total assets). It is an ex ante measure of expected losses from lending activities (Berger et al., 2010). Baele et al. (2007) report that banks with a high loan loss provision ratio exhibit higher risk. In a different line of arguments, Lee et al. (2014) find a positive impact of loan loss provision ratio on bank profitability as large profit margins are generally accompanied with greater risks. Thus, the impact of this ratio is unknown à priori.

Since there is wide consensus that countries with growing and prosperous economies and enjoying greater financial development have the most stable banks (Nguyen et al., 2012b), we control for financial development. For this purpose, we proxy it by the annual market capitalization to gross domestic product. It is expected that, as the financial sector flourishes, banks have high propensity to boost their performance. Sometimes, banks that operate under a stringent banking environment may be inclined to mix banking and nonbanking financial activities. It goes in line with the findings of Ammar and Boughrara (2019b). Furthermore, removing restrictions on banks' operations has a positive impact on banking firms' shift toward nontraditional activities. The freest banking sectors are filled with banks that efficiently choose the best investments that generate higher profit margins. It is however expected that the increased consolidation of financial activities might not be restrained by the weakness of the financial markets. This result corroborates those of Ammar and Boughrara (2019b). Now and then, there is no synchronization between the requirements of a more open banking sector that allows diversification with the state of the financial markets. To account for this, we add to the control variable set another country-specific variable, namely an index measuring banking freedom. It is a country-specific annual index of financial freedom (the index is graded using a scale from 0 to 100, where 100 represents maximal freedom).

Finally, we control for the main variables that account for the business cycle. We insert the macroeconomic variable, the gross domestic product growth, which stands for the annual real gross domestic product growth and is an external determinant of macroeconomic conditions. The gross domestic product growth is also highlighted as a driving force for the diversification decision. In upturns, banks diversify their revenue streams to widen their profit margins and to construct "reserve profit revenue" as a cushion against shocks when the demand for credits shrinks in downturns. Nontraditional banking products become important during recession (Nguyen et al., 2012a). Likewise, we incorporate another macroeconomic variable which is the inflation rate. The impact of the annual consumer price inflation on bank performance depends on whether the former is anticipated or unanticipated (Perry, 1992). A positive impact might appear when banks timely adjust their interest rates. However, a negative effect rises when banks show slowness in adjusting their interest rates which in turn results in a fast growth of bank costs compared to revenues. Hence, the impact cannot be determined à priori. The country-specific macroeconomic variables and banking industry data are retrieved from the World Development Indicators database (World Bank, 2020). The index of banking freedom is obtained from the Heritage Foundation (2020). The variables used to estimate the impact of revenue diversification on MENA banks' performance are constructed based on prior literature, the Bankscope, the World Bank (world development indicators' database) and the Heritage Foundation (see Appendix A).

It is worth noting that the key variable we aim to assess the impact on bank performance is the degree of noninterest income mix, div_non (see Appendix A). The distribution of the noninterest income components per country and for both subsamples (GCC and non-GCC)[3] shows that while the distributions of noninterest income components appear to be dominated by commissions for all MENA banks, Trade seems to gain important shares when compared with other noninterest revenue when it comes to GCC-based banks. The picture is somewhat different for non GCC-based banks insofar as the distributions appear less homogenous. Globally, the information conveyed by the noninterest income distribution appear in line with the related literature that more developed countries banks invest more in trading activities but in developing countries noninterest activities are essentially driven by fees and commissions.

4. GPS-DRF setup and implementation of estimation strategy

From a policy evaluation perspective and in many settings, it is not relevant to consider the treatment as binary. However, the focus should be on the level of exposure (or dose) provided. In our case, banks undertake a diversification strategy, but each banking firm has a degree of exposure to nontraditional activities. As the revenue diversification variable (*div_non*) takes on a continuum of values, lying in the interval $]0,1[$,[4] we follow Imbens (2000), Hirano and Imbens (2005) and Bia and Mattei (2008) and use the *GPS* method[5] to assess the impact of engaging in nontraditional business lines on the performance of MENA banks. Since the *GPS* approach has the merit to better suit the real world conditions, as in social science in general and in diversification context in specific, bank noninterest revenue mix takes on a continuum of values (Imbens & Wooldridge, 2009; Imbens & Rubin, 2015). The *GPS* addresses the confounding bias that occurs in observational studies. It assigns an observation to either the treatment (diversified MENA banks) or the control (concentrated

[3] Details are available from the corresponding author on request.

[4] *div_non* is a Herfindahl Hirschmann concentration index (HHI) where its values lie in the interval τ. See Appendix A for a detailed definition of *div_non* measure.

[5] It is worth noting that the generalization of the well established and widely used PSM methodology for binary treatments (Rosenbaum & Rubin, 1983) for the case of continuous treatments has been developed according to two slightly different approaches or methodologies. The first one, which hinges on the so-called *GPS*, was suggested by Hirano and Imbens (2005), while the second one, which is based on the P-function, was developed by Imai and A van Dyk (2004). While the two approaches appear similar in spirit, the Hirano and Imbens (2005) approach has a theoretical advantage over the Imai and A van Dyk (2004) method in that the former can be used to estimate the full dose–response function (*DRF*) rather than the simple average treatment effect, which is often estimated with the Imai and A van Dyk (2004) method.

MENA banks) group, conditional to observed prediversification strategy covariates. In the absence of experimental data, this appealing empirical tool can mimic a randomized controlled trial ex post. This extension of the classical PSM models (of binary treatment variables) allows us to estimate the impact of shifting toward off-balance sheet activities on bank profitability and risk, arising from variations in treatment levels (i.e., different levels of exposure to nontraditional activities). Focusing on the dynamics of the degree of exposure of a bank to off-balance sheet activities amounts to estimate a DRF. This method has been extensively applied in the economic literature (among others Fryges, 2009; Kluve, Schneider, Uhlendorff, & Zhao, 2012). Still, it has not become a tool of analysis in other research areas such as finance. Hence, we deem this study as the first empirical investigation that examines the bank performance–noninterest revenue mix nexus using the *GPS* approach.

4.1. Model setup

The empirical setup involves a modeling of the *GPS* as well as an estimation of the conditional expectation of the outcome of interest, the so-called *DRF*, as a joint function of the estimated *GPS* and the income diversification proxy. For this study, the main goal is to estimate the *DRF* and assess the impact of different income diversification exposures on MENA banks risk-adjusted profitability (i.e., raroaa and raroae) and financial stability (i.e., z-score). It is worth noting that our empirical setup relies on work of, among others, Hirano and Imbens (2005) and Imbens (2000).

Let D denote the diversification variable, d be a specific degree of noninterest income mix lying in the interval τ, such that $\tau \subseteq]0,1[$, and $Y_i(d)$ a random variable of a particular potential treatment d to a potential outcome. In our study, d stands for the potential impact on bank performance corresponding to the specific level of diversification received/adopted. It is worth noting that the latter is measured by the HHI index, which we normalize in order to rescale it to always range between 0 and 1. Let D_i denote the treatment variable which represents the degree of exposure to diversification pursued by bank i. D_i is, therefore, a continuous random variable whose realizations may theoretically be any fractional value d.[6] Let $gps(d,X)$ denote the conditional density of the continuous degree of exposure to diversification D given the covariates X and computed as:

$$gps\Big(d,X\Big) = gps_{D|X}\Big(d,X\Big) \tag{1}$$

Let $GPS_i = gps(D_i,X_i)$ denote the conditional density at the treatment level actually received by bank i, and let $GPS_i^d = gps(d,X_i)$ denote the family of random variables indexed by d. Clearly, for banks with $D_i = d$, we have $GPS_i = GPS_i^d$ where:

$$GPS(d,X) = gps(D_i = d, X_i) \tag{2}$$

The objective is to estimate the outcome of bank i at each level of the treatment, the so-called *DRF*. When averaging the outcome over the entire set of banks $i \in \{1,\ldots,N\}$, we obtain the average *DRF*:

$$\mu(d) = \mathbb{E}(Y(d)) \tag{3}$$

It is also interesting to compute the pairwise treatment effects of the following form:

$$\mathbb{E}\left(\Delta^{d1,d2}\right) = E[Y(d_1) - Y(d_2)] \tag{4}$$

which, when d_1 tends to d_2, can be interpreted as derivatives indicating the growth effects induced by a marginal increase in the diversification level from d_1 to d_2.

4.2. Covariates used

The covariates are (all) the prediversification take-up factors that have an impact on bank profitability, and the decision to implement a diversification strategy (see Appendix A). The factors may be split into bank-specific factors such as the bank size, the capital ratio, the liquidity ratio, the loan loss provision ratio, the deposits-to-total-assets ratio, the net interest margin ratio, the bank growth and the bank/management efficiency, and into industry-specific characteristics, including[7] gross domestic product growth, inflation, banking freedom and financial development. The selection of the factors is based on the empirical literature as regards revenue diversification-bank performance nexus.

4.3. Assumptions

For efficient estimation of the *DRF*, three assumptions must hold (Robins, 1997; Imai & A van Dyk, 2004; Imbens, 2000). Firstly, $Y_i(d)$, which is the outcome of the diversified bank, is not affected by what kind of noninterest revenue mix is adopted or by how the diversification decision is undertaken by the other MENA banking firms. This assumption is labeled as the *stable unit treatment value*

[6] In the light of our bank sample, d is bounded between d_{min} and d_{max} such that $d \in \tau = [d_{min}, d_{max}] \subset]0,1[$. Such a restriction will condition the choice of the probability density function of the treatment.

[7] Data details and covariates construction were presented in 3.2 subsection.

assumption (Imbens & Rubin, 2015). Secondly, the *GPS* relies on the weak unconfoundedness assumption (or the so-called Conditional Independence Assumption (CIA)) to adjust for prediversification decision differences between diversified and concentrated MENA banks. This assumption (Hirano & Imbens, 2005) requires the conditional independence to hold for each value of the degree of the shift towards nontraditional activities rather than the joint independence of all potential outcomes as follows: D is weakly unconfounded, given X, if $Y_i(d) \perp D|X$ for all realizations $d(d \in \tau)$ of the treatment D, where $Y(d)$ is the outcome associated with the exposure level d. Else, $Y_i(d) \perp D|X \forall d \in \tau$, where X stands for bank-specific and industry-related variables, D is the diversification variable, d is a specific degree of noninterest income mix lying in the interval τ, and $Y(d)$ is the potential impact on bank performance corresponding to the specific level of diversification pursued. Lastly, banks within the same strata of the *GPS* have to look identical in terms of their observed characteristics and independently of their level of exposure to diversification insofar as $X \perp \mathbb{1}_{\{D=d\}} \big| gps(d,x)$, where $\mathbb{1}_{(.)}$ is an indicator function. Combined with the weak unconfoundedness assumption, the balancing property implies that:

$$f_D(d|gps(d,X_i), Y_i(d)) = f_D(d|gps(d,X_i)) \tag{5}$$

Hence, the validity of the *GPS-DRF* method is crucially based on the assumptions of the independence of the outcomes (i.e., raroaa, raroae, and z-score) and the dose of the treatment (i.e., degree of revenue diversification) as well as on the balancing distributions of bank- and industry-specific factors.

4.4. Implementing GPS-DRF approach

The first step of the *GPS-DRF* implementation process evolves the estimation of the score. For this purpose, the choice of the relevant specification of the treatment equation is crucial. The specification suggested by Hirano and Imbens (2005) and developed by Bia and Mattei (2008) requires a strict assumption to be verified, namely the distribution of the treatment given the preprogram covariates has to be normal. In many cases, it is difficult to meet this strict condition. This version of the *GPS-DRF* approach specification cannot be implemented whenever the distribution of the treatment in question (or its transformation) deviates from the normal distribution. As a consequence, the causal effect literature proposed a new approach that takes into account the possibility of failure to meet the normality assumption (Imbens & Wooldridge, 2009; Imbens & Rubin, 2015). In fact, Guardabascio and Ventura (2014) opted for a more general and flexible approach by using the Generalized Linear Model (hereinafter, *GLM*) to estimate the *GPS* rather than the strictly linear regression. Indeed, the advantage of the *GLM* is that it constitutes a general modeling framework that allows the response variable (in our case, $\boldsymbol{D}$) to have an error probability distribution other than the normal one and by allowing the linear model (i.e., $\boldsymbol{\beta}'\mathbf{X}$) to be related to the response variable via a relevant link function. By considering various link functions and different probability distribution, the *GLM* framework encompasses a rich set of models (de Jong & Heller, 2008; Dobson & Barnett, 2018). The proposed distribution families are Poisson, binomial, normal, negative binomial, inverse Gaussian, and gamma. The choice of the distribution families as well as their corresponding links is driven by the nature of the treatment proxy. Following the *GLM* and the treatment effect literature (see, among others, Nelder & Wedderburn, 1972; de Jong & Heller, 2008; Bia & Mattei, 2008; Bia & Mattei, 2012; Guardabascio & Ventura, 2014), we assume that the distribution of $\boldsymbol{D}$ is specified from the exponential family as follows:

$$f(D) = \boldsymbol{h}(D,\phi)\exp\left(\frac{b(\theta)D - a(\theta)}{\phi}\right) \tag{6}$$

where $a(\theta)$ refers to the family, and θ and ϕ are the canonical and dispersion parameters, respectively. When $b(\theta) = \theta$, then the distribution takes its canonical form.[8] In addition to that, if ϕ is known, then θ is called the canonical (standard) parameter and is related to the mean of the distribution through:

$$g[\mathbb{E}(D)] = \dot{a}(\theta) = \boldsymbol{\beta}'\mathbf{X} \tag{7}$$

where $g(.)$ is the link function—a monotonic and a differentiable function—determining the relationship between the mean and the covariates $X, \dot{a}(\theta)$ is the first derivative of $a(\theta)$ with respect to θ and β is $(k \times 1)$ vector of parameters. Since our treatment variable D (the noninterest income mix) belongs to the unity interval and hence it is fractional, the impact of any bank- and industry-specific covariate cannot be constant over its range, as stressed by Guardabascio and Ventura (2014). However, some methodological issues arise with fractional dependent variables, especially in the linear context modeling. Thus, the drawbacks of the linear models for fractional data are analogous to the drawbacks of the linear probability model for binary data. One way to mitigate these drawbacks is to consider the link function $g(.)$ in Eq. (7) as the log-odds function (see, among others, Papke & Wooldridge, 1996), so we have:

$$log\left[\frac{D_i}{1-D_i}\right] = \boldsymbol{\beta}'\mathbf{X_i} + \mathbf{u_i} \tag{8}$$

The appeal of Eq. (8) is that its left-hand side can take on any real value when $D_i = d \in \tau = [d_{min}, d_{max}] \subset]0,1[$, and therefore stands well defined. As appealing as it may seem, Eq. (8) yields the parameters of interest difficult to interpret in our context. Again, it is still hard

[8] It is always possible to express any probability distribution in its canonical form by re-expressing θ as θ' and performing the change $\theta = b(\theta')$. For further details on *GLM* modeling, readers may consult de Jong and Heller (2008).

to compute $\mathbb{E}[D_i|\mathbf{X_i}]$ without further assumptions. Indeed,

$$\mathbb{E}[D|\mathbf{X}] = \int_{-\infty}^{\infty} \frac{exp(\boldsymbol{\beta}'\mathbf{X}+\mathbf{u})}{1+exp(\boldsymbol{\beta}'\mathbf{X}+\mathbf{u})} f_{\boldsymbol{u}}(\omega|X)d\omega \tag{9}$$

where $f_{\boldsymbol{u}}(.|X)$ is the conditional density of $\mathbf{u}$ in Eq. (8) and ω is a dummy argument of integration. Even if we assume that the covariates are exogenous[9] (i.e., $\mathbf{u}\perp\mathbf{X}$), $\mathbb{E}[D|\mathbf{X}] \neq \frac{exp(\boldsymbol{\beta}'\mathbf{X})}{1+exp(\boldsymbol{\beta}'\mathbf{X})}$.[10] When $\mathbf{u_i}$ is not orthogonal to $\mathbf{X}$, Eq. (9) cannot be estimated without estimating the density $f_{\boldsymbol{u}}(.|X)$. To sidestep this difficulty, Papke and Wooldridge (1996) recommended directly specifying models for $D_i|\mathbf{X_i}$ without having to estimate $f_{\boldsymbol{u}}(.|X)$.[11] In particular, the authors suggested assuming:

$$\mathbb{E}(D) = \boldsymbol{F}(\boldsymbol{\beta}'\mathbf{X}) \tag{10}$$

where $0 < \boldsymbol{F}(s) < 1 \forall s \in \mathbb{R}$, which assures that the predicted values of D fall in]0,1[. By and large, cumulative distribution functions (cdf) satisfy such conditions, and one special attractive cdf is the logistic function where $\boldsymbol{F}(s) = \Lambda(s) = [1+exp(-s)]^{-1}$. Under the above assumption (that is Eq. (10)), it is possible to estimate the parameters of interest $\boldsymbol{\beta}$ by having recourse to the Quasi-Maximum Likelihood (QML) procedure, which yields a consistent and $\sqrt{N}$-asymptotically normal estimator of the vector of the parameters of interest, $\mathbb{l}(\beta)$, whatever the distribution of $\boldsymbol{D}|\boldsymbol{X}$. The expression of the log QML (or, shortly, the Bernoulli log-likelihood) function is expressed as:

$$\mathbb{l}\left(\beta\right) = \sum_{i=1}^{N}[D_i log\{\Lambda(\boldsymbol{\beta}'X_i)\} + (1-D_i) log\{1-\Lambda(\boldsymbol{\beta}'X_i)\}] \tag{11}$$

Once $\widehat{\beta}$, which is the vector of parameters of interest, is retrieved from the previous step (see Eq. (11)), we follow Guardabascio and Ventura (2014) and start by estimating the parameters θ and ϕ of the selected conditional distribution of the diversification variable given the pre-diversification covariates. Afterward, the *GPS* can be estimated based on the Bernoulli log-likelihood function for each bank i as follows:

$$\widehat{GPS}_i = [\Lambda(\widehat{\boldsymbol{\beta}}'\mathbf{X}_i)]^{D_i}[\Lambda(1-\widehat{\boldsymbol{\beta}}'\mathbf{X}_i)]^{(1-D_i)} \ \forall i \in \left\{1,\ldots,N\right\} \tag{12}$$

Subsequent to the estimation of the *GPS*, the verification of the balancing property is required. This is an important assumption that the continuous treatment exposure method shares with the mainstream *PSM* models.[12] As stressed by Imbens (2000), adjusting for the *GPS* removes all the bias associated to the differences in preexposure variables (in our case, the bank- and industry-specific factors) between diversified MENA banks and their focused counterparts. For this purpose, we split the diversification variable range into three (quasi) equally spaced intervals. Then, we calculate the median of the diversification exposure and evaluate the *GPS* at that specific level of D, $gps\left(\widetilde{D}_j, x\right)$, where $\widetilde{D}$ is the median of the diversification degree among banks with $\{d_j < D \leqslant d_{j+1}\}$ (within each diversification interval). For each diversification interval, we split the estimated *GPS* range into five equally spaced intervals. Within each *GPS* interval and for each covariate, we compute the difference between the mean of each covariate among banks with $\{d_j < D \leqslant d_{j+1}\} = 0$ against banks with $\{d_j < D \leqslant d_{j+1}\} = 1$. Next, we combine differences in means, weighted by the number of observations in each group of the *GPS* interval and for each diversification interval. The procedure is repeated for each $d_1, \ldots, d_j$ that covers the range of observed D. Then, we compare the extreme value of the Bayes factors test statistics[13] for equality in means (its lowest value) with reference values to know to what extent the balancing property is supported by the data.

Relying on the *GPS*, we use the following Gaussian quadratic regression model to estimate the expectation of the outcome conditional on the diversification levels and on the (estimated) *GPS* as:

$$\mathbb{E}[Y|D, \widehat{GPS}] = \alpha_0 + \alpha_1.D + \alpha_2.D^2 + \alpha_3.\widehat{GPS} + \alpha_4.\widehat{GPS}^2 + \alpha_5.D \times \widehat{GPS} \tag{13}$$

[9] Regarding this point, Wooldridge (2010) suggested assuming that the log-odds transformation yields a linear model with an additive error, where *ipso facto* entailed that $X \perp u$.

[10] It is nonetheless still possible to estimate it using non-parametric methods, such as the smearing method (see Duan, 1983), which is less demanding than the fully parametric methods. However, such methods have not allowed computing the partial effects on $\mathbb{E}(D|X)$ to which we are interested.

[11] This is either difficult or not robust depending on whether a non-parametric or a parametric specification is adopted.

[12] It has been documented that the *GPS* has a balancing property similar to the balancing property of the propensity score for binary treatments. Put simply, within strata with the same value $g(d,X)$, the probability that $D = d$ does not depend on the value of X; i.e., the *GPS* has the property that $X \perp \mathbb{1}_{\{D=d\}} \left| gps(d,x)\right.$. According to Hirano and Imbens (2005), this is an *ipso facto* implication of the *GPS* definition and does not call for unconfoundedness. In addition, Hirano and Imbens (2005) demonstrated that, if the assignment to treatment was weakly unconfounded given covariates X, then it was also weakly unconfounded given the *GPS*.

[13] The Bayes factor is a Bayesian hypothesis testing and model selection criterion introduced by Jeffreys (1961).

Table 1
Estimation of the GPS (Whole sample).

VARIABLES	Whole sample
size	−0.040***
capitalratio	0.006***
liquidity	−0.043
dta	−0.053
bank_growth	−0.000
bank_efficiency	0.029*
llp_r	0.032
inf	0.001
findev	−0.002***
Constant	0.606***
Log Pseudolikelihood	−638.358
AIC	0.949
Observations	1367

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

The quadratic form[14] (i.e., second-order polynomials of the diversification measure) is applied to the model to account for a potential nonlinear nexus between income diversification and MENA banks performance. Following Hirano and Imbens (2005), Eq. (13) is estimated using the OLS method.[15] The last step focuses on the estimation of the continuous *DRF*, defined as the average expected MENA banks' performance at a specific diversification level d and using the estimated coefficients, namely $\{\widehat{\alpha_0}, \ldots, \widehat{\alpha_5}\}$, from Eq. (13) as follows:

$$\mathbb{E}\left[\widehat{Y(d)}\right] = \frac{1}{N}\sum_{i=1}^{N}\left(\widehat{\alpha}_0 + \widehat{\alpha}_1.D + \widehat{\alpha}_2.D^2 + \widehat{\alpha}_3.\widehat{gps}\left(d,X_i\right) + \widehat{\alpha}_4.\widehat{gps}(d,X_i)^2 + \widehat{\alpha}_5.D \times \widehat{gps}\left(d,X_i\right)\right) \tag{14}$$

where N stands for the total number of banks in the data sample, $\mathbb{E}[\widehat{\mathbb{Y}(\mathbb{d})}]$ is a dose-specific score (i.e., an average response evaluated had all banks received a particular dose of diversification d) and the estimated *gps* for each bank i is evaluated at $D_i = d$ and not at D; i.e., $\widehat{gps}(d,X_i) = \widehat{gps}(D_i = d,X_i)$.[16]

To obtain an estimate of the entire *DRF*, Eq. (14) is calculated for the entire range of the diversification measure (and by increasing its degree by 0.05 in each step). The confidence intervals of the *DRF* are determined via bootstrapping. The last step of the *DRF-GPS* methodology consists in plotting the estimated *DRF*s to get the predicted MENA banks performance, conditional to bank- and industry-specific pretreatment covariates, for each level of diversification.

5. Empirical results

The parameter estimates of the GLM approach (for the *GPS* model) are reported in Table 1 for the whole MENA sample. As mentioned in the previous section, the *GPS* is computed as the conditional density of the income diversification measure on pre-diversification covariates. The results reported in Table 1 reveal that the income diversification proxy is driven by the bank size, the capital ratio, the bank efficiency and the degree of the financial development. The findings demonstrate that large MENA banks are steering away from revenue diversification. Yet, their highly capitalized peers exhibit a tendency to increase their interest income mix. Surprisingly, banks that do not effectively manage their operating costs tend to extend their product lines. Under unfavorable conditions (i.e., low degree of financial development), MENA banks prefer focusing on a specific source of noninterest income to engage in different market segments that simultaneously produce fees, commissions, trading revenue, and other noninterest income.

As the soundness of the *DRF*-based inference depends on the estimation quality of the *GPS*, we evaluate the validity of the latter through the balancing property test.[17] To this end, we divide the range of revenue diversification into three treatment intervals. More specifically, we define the diversification intervals as [0.333, 0.399], [0.399, 0.699] and [0.699, 0.900]. The results of the test, which are reported in Table 2, clearly show that the balancing property of the baseline covariates is satisfied, according to the Bayes factor.

The last step consists in estimating the *DRF*. Table 3 includes the estimation results of the conditional expectation(s) of the outcome

[14] We follow Bia and Mattei (2008) and opt for the quadratic form.

[15] As outlined by Hirano and Imbens (2005), the estimated coefficients $\{\widehat{a_i}, i = 1, \ldots, 5\}$ do not have any trivial sense and thus are not economically interpretable. They rather correspond to the *DRF* of diversification level d, which has a causal interpretation if compared with another diversification level, say d' (see, among others, Kluve et al., 2012).

[16] D is a random variable, and d is a realization of D.

[17] This entails assessing the degree to which conditioning on the propensity score (via matching, weighting, or stratification) has balanced the measured baseline covariates between focused banks and their diversified counterparts.

Table 2
Covariate balancing property check (Whole sample).

	TG No. 1: [0.333, 0.399]	TG No. 2: [0.399, 0.699]	TG No. 3: [0.699, 0.900]
size	2.667	6.105	5.468
capitalratio	4.879	5.997	5.930
liquidity	1.640	3.605	5.874
dta	4.797	6.299	5.507
bank_growth	1.224	1.712	5.462
bank_efficiency	2.977	5.308	5.021
llp_r	1.038	1.433	4.418
inf	5.023	6.292	5.458
findev	1.994	6.420	1.357

Notes: TG stands for the treatment group. The Bayes factor is a test statistic for equality in means. The Bayes factors are interpreted following Jeffreys (1961, Appendix B, p. 432–433) as the following: Let $B01$ denote the Bayes factor for the nested models $M_0 \subset M_1$, then $B_{01} > 1$: evidence supports M_0; $1 > B_{01} > 10^{-0.5}$: very slight against M_0; $10^{-0.5} > B_{01} > 10^{-1}$: moderate against M_0; $10^{-1} > B_{01} > 10^{-2}$: strong to very strong against M_0; $10^{-2} > B_{01}$: decisive against M_0. According to the Bayes factor, the balancing property is verified.

Table 3
Estimates of the conditional expectation(s) of the outcome(s)—Estimated DRFs.

Covariates	raroaa	raroae	z-score
div_non	−11.774	−10.002	−604.908**
div_non_sq	6.404*	7.609***	170.633*
pscore	−113.561***	−62.385*	−509.832
pscore_sq	79.258**	42.013	14.793
div_non_pscore	8.832	0.979	761.261
Constant	42.941***	26.894***	358.857
Observations	1367	1367	1367

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

(s) raroaa, raroae, and z-score. We estimate the *DRF*s over the entire range of diversification levels (by increasing its degree by 0.05 in each step). The estimated *DRF*s show the predicted MENA banks performance, conditional to bank and industry-specific pre-diversification factors, at each level of revenue mix. The resulting *DRF*s relative to raroaa, raroae, and z-score are depicted in Figs. E.1–E.3, respectively (see Appendix E).[18] Standard errors are computed by the bootstrapping technique from 100 replications. Along with the predicted *DRF*s, the estimated derivatives are also displayed for the[19] three bank performance measures.

For all the figures (see Appendix C, Appendix D, and Appendix E), the blue line shows the predicted GCC, non-GCC and MENA banks' performance while the green and red lines indicate the 95% confidence bounds. The *DRF*s associated to the three samples (whole, GCC, and non-GCC) exhibit different shapes. When considering the whole sample, it stands out from Appendix E that the *DRF*s associated to the different key variables depict a U-shaped curve with more or less similar characteristics. On average, banks performance, as measured by raroaa reaches its minimum (lowest level of 2.60%) when the treatment (diversification) does attain the 50% threshold level. Afterward, it builds up progressively as diversification moves away from 50%. The same analysis holds when banks' performance is measured by raroae, except that the lowest performance recorded at the 62% noninterest revenue mix level approximates 2.2%.

When it comes to the *DRF* associated with the z-score, we notice that while it exhibits a roughly similar pattern as those associated with both raroaa and raroae, it nonetheless depicts some different characteristics. Indeed, after reaching its lowest level of 33% at the 50% diversification level, it goes up smoothly as the revenue concentration increases toward the 60% level where it records a value of 50%. Starting from the 60% level onward, it rises smoothly until exceeding 50% when the noninterest income concentration proxy approaches the 100%.

Our sample is heterogeneous as it includes the set of banks in the GCC countries and their peers established in the rest of the MENA region. We split our sample into two groups, namely GCC and non-GCC-based banks since the banking firms of the former operate in more developed banking sectors, perform in more developed financial systems, benefit from higher degrees of banking freedom, and have different business practices and specifically different perceptions of corporate behavior vis-à-vis revenue diversification decision compared with their counterparts established in non-GCC countries.

[18] Following Bia and Mattei (2012), the graphs of the DRFs also accounted for treatment values in the interval]0, 0.333[.

[19] While our sample extends to 2020, it was not our intention to appraise the impact of Covid-19 pandemic on the propensity of MENA banks to shift toward nontraditional activities. We keep this promising line of research for the future. Besides, employing low frequency data (i.e., yearly) does not allow to unveil any significant impact of the pandemic as removing banks 2020 observations did not affect the inference (results were not reported but are available upon request from the corresponding author).

Table 4
Estimation of the GPS(GCC non-GCC subsamples).

Covariates	Non-GCC subsample	GCC subsample
size	−0.352***	−0.022**
SIZE_SQ	0.018***	
capitalratio	0.001	0.002**
liquidity	−0.074	−0.039
dta	−0.046	0.038
nim	0.069***	0.000
bank_efficiency	0.746*	
llp_r	0.011	0.425**
bank_growth	0.001	0.000
gdp_growth	−0.013**	0.003
bankingfr	−0.017**	−0.000
findev	−0.001***	
Constant	2.743***	0.268
Log Pseudolikelihood	−367.549	−621.696
AIC	0.957	0.948
Observations	795	1333

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table 5
Covariate balancing property check (Bayes factor).

Covariates	TG No. 1: [0.333, 0.399]		TG No. 2: [0.399, 0.699]		TG No. 3: [0.699,0.900]	
	GCC	Non-GCC	GCC	Non-GCC	GCC	Non-GCC
size	7.251	3.030	6.794	5.246	3.337	1.411
SIZE_SQ		3.258		5.290		1.283
capitalratio	3.968	3.791	4.025	5.821	2.748	3.503
liquidity	4.974	1.029	7.442	1.472	7.563	3.971
dta	5.949	3.195	6.299	2.747	5.810	1.744
nim	6.413	2.952	9.741	4.645	4.400	4.585
bank efficiency		2.046		5.445		2.255
llp_r	4.467	1.136	6.626	1.522	4.720	2.883
bank_growth	1.338	3.624	3.816	5.931	7.020	5.275
gdp_growth	1.866	3.243	2.181	5.788	3.213	3.826
bankingfr	7.086	0.349	6.871	0.351	5.526	4.524
findev		1.677		6.001		1.393

Notes: TG stands for the treatment group. The Bayes factor is a test statistic for equality in means. The Bayes factors are interpreted following Jeffreys (1961, Appendix B, p. 432–433) as the following: Let $B01$ denote the Bayes factor for the nested models $M_0 \subset M_1$, then $B_{01} > 1$: evidence supports M_0; $1 > B_{01} > 10^{-0.5}$: very slight against M_0; $10^{-0.5} > B_{01} > 10^{-1}$: moderate against M_0; $10^{-1} > B_{01} > 10^{-2}$: strong to very strong against M_0; $10^{-2} > B_{01}$: decisive against M_0. According to the Bayes factor, the balancing property is verified.

The parameter estimates of the *GPS* for the subsamples of the GCC and the non-GCC are reported in Table 4. It shows that the bank size, the capital ratio and the loan loss provision ratio exert a major influence over the diversification of revenue streams decision for the GCC-based banks while the bank size, its squared term, the net interest margin ratio, the bank efficiency, the GDP growth, the banking freedom and the financial development stand as the main drivers of the increasing shifts towards nontraditional activities for their non-GCC peers. Basically, large GCC and non-GCC banks are less inclined to diversify their revenue streams. In fact, they have more incentives to avoid the product expansion strategy as their organizational complexity, resulting in potential agency problems between managers and shareholders, which may increase the risks incurred by these institutions. Moreover, the managers who seek to achieve their own interests are more prone to undertake myopic and value-decreasing investment strategies at the expense of lower performance. Nevertheless, the impact of size seems to be exhausted at a certain level for non-GCC banks. The attitude of bank managers toward the diversification decision drastically changes once they attain a certain level of managerial expertise, technology investments, and human capital, so they try to explore new market segments in search for higher performance. This drastic change may also be explained by the wish of non-GCC banks to benefit from the too-big-to-fail protection while increasing their market shares and becoming multi-product financial institutions. Highly capitalized GCC banking firms have a high propensity to engage in nontraditional activities as higher shares of capital shelter them from potential unexpected market shocks. Furthermore, banking firms with high credit risk have more incentives to shift toward off-balance sheet activities to compensate for losses in loan-making business lines, lending support to Baele et al. (2007).

In addition, non-GCC banks that are successful in their loan-making activities are eager to reach new market segments in search of higher profits and better financial stability. The results also show that non-GCC banking firms that do not efficiently manage their

Table 6
Estimates of the conditional expectation(s) of the outcome(s)—Estimated DRFs.

Covariates	Panel A: Non-GCC subsample outcomes			Panel B: GCC subsample outcomes		
	raroaa	raroae	z-score	raroaa	raroae	z-score
div_non	−20.809***	−18.769***	−904.133***	−11.770	−12.619	−231.741
div_non_sq	1.237	8.717**	83.130	−3.932*	−5.486**	−48.115
pscore	19.151	15.465	271.911	21.386	74.260	−4277.215***
pscore_sq	−35.519*	−22.034	−892.777*	−50.879	−98.784	3763.465***
div_non_pscore	35.482**	13.503	1443.742***	28.491	33.217	521.625*
Constant	3.183	4.107	188.286	4.631	−10.471	1219.858***
Observations	795	795	795	1331	1331	1331

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

operating costs have more incentives to shift toward noninterest income generating activities. This finding goes against prior evidences (Elsas et al., 2010; Nguyen et al., 2012b). In downturns, when the demand for credits shrinks, non-GCC banks shift towards off-balance sheet activities to compensate for and withstand the losses incurred from loan-making business lines. Our finding echoes those of Nguyen et al. (2012a) who documented that shifts toward noninterest revenue-generating products and services would go up during recession periods. More than that, the adverse macroeconomic and industry-specific conditions (underdeveloped financial markets and banking sectors with a low banking freedom) do not seem to impede non-GCC banks from widening the scope of their business activities. This corroborates the findings of Ammar and Boughrara (2019b) that showed that non-GCC banks would diversify their revenue streams under unfavorable market conditions.

Again, the results of Table 5 reveal that, after *GPS* adjustment, differences in prediversification covariates are mitigated for both GCC and non-GCC subsamples. Panels A and B of Table 6 display the estimates of the conditional expectation(s) of the outcome(s) raroaa, raroae, and z-score for the non-GCC and GCC subsamples, respectively.[20] In the same vein as Table 3, the Table 6 estimates refer to Eq. (14) of the *GPS* procedure.

Plots of the *DRF* curves for the predicted risk-adjusted profitability raroaa, raroae and financial stability z-score are reported in Figs. C.1–C.3 for the GCC subsample, respectively (see Appendix C) and and Figs. D.1–D.3 for the non-GCC subsample, respectively (see Appendix D).

A close inspection of *DRF*s associated with the GCC subsample (see Appendix C) indicates that they completely differ from those depicted in Appendix D. Hence, they induce different conclusions regarding bank behaviors. Chiefly, the *DRF*s display an inverse U-shaped curve. When focusing on the *DRF* of the raroaa indicator, we notice that the bank performance culminates at approximately a 50% diversification level. Then, it gradually reduces in an inverse proportion to the diversification growth. However, when a noninterest income mix concentration level exceeds 60%, the banks performance, as measured by raroaa indicator, tends to decline slowly and monotonously.

The same analysis holds when bank risk-adjusted profitability is proxied by raroae. When focusing on the *DRF* associated with the insolvency risk measure z-score, we notice that while it exhibits a roughly similar pattern as those associated with both raroaa and raroae, it nonetheless depicts some different characteristics. Indeed, after reaching its highest level of 21% at the 50% diversification level and starting from the 70% income mix level onward, it shrinks gradually until attaining the 15% level when the noninterest income concentration proxy approaches the 100%. Unlike previous *DRF*s (i.e., associated to GCC subsample), the *DRF*s associated with the non-GCC subsample is rather U-shaped (see Appendix D) indicating that the bank performance records its minimum value for a diversification level around 50% (2.5% for raroaa and 35% for z-score). When the noninterest revenue concentration intensifies to the level of 62%, the banks performance (i.e., raroae) attains its lowest level of 2.1%. Beyond these diversification thresholds, the non-GCC bank performance peaks up again.

Our paper findings reveal that the optimal diversification level for the GCC banks is located around the value of div_non = 0.50, thereby corroborating what the empirical literature has documentated regarding the reference (threshold) level that guarantees a perfect mix of noninterest income streams (Ammar & Boughrara, 2019b). It is the diversification level that allows the GCC banking firms to reap the benefits in the form of higher profits and more financial sustainability. However, this threshold value (of 50% for raroaa and z-score, and of 62% for raroae) leads to the lowest performance of their non-GCC peers. Hence, a reliance on interest income generation or the focus on a specific noninterest income source appears as a sound strategy to restore profitability and to mitigate the insolvency risk for the non-GCC banks.

By and large, the visual inspection of the graphs shows disparities between the GCC and the non-GCC banks. Starting with the GCC subsample (see Appendix C), a value of div_non = 0.5 of noninterest income mix appears beneficial for GCC banks insofar as it enhances their performance. Exceeding this threshold level of div_non (hence more concentration), tends to exert an adverse (i.e.,

[20] It should be of no surprise that the parameters α may exhibit high values insofar as they have no direct interpretation as stressed by Hirano and Imbens (2005). Again, such high values have been documented by, among others, Kluve et al. (2012),uardabascio and Ventura (2014) and Neves, Silva, Freitas, and Braga (2021). Their statistical significance can be deemed as evidence of the existence of the bias brought about by the characteristics considered (Hirano & Imbens, 2005).

negative) impact on GCC banks' profits and financial stability (the decreasing trend above the 0.5 level). Reducing the noninterest income mix by focusing on one component (whether its fees and commissions, trading-related activities, or other noninterest income business lines) or focusing on the mainstream loan-making activities does not seem to be the best strategy to attain the greatest performance. Ergo, the best option (in terms of higher risk-adjusted profitability and financial stability) is to reach the maximum diversification level of 0.5. In fact, the GCC banks operating in developed financial markets have more access to higher technologies and benefit from higher banking freedom and hence, are the best placed to take advantage of their existing capabilities and diversify their revenue streams. Hence, the GCC banking firms that expand toward nontraditional activities (seeking higher profits) seem to opt for the best scenario that favors stable performance. This finding is in line with those reported by Abuzayed et al. (2018) who documented greater financial stability for GCC banks at higher levels of diversification.

Still, the non-GCC banks that focus on their traditional mainstream business lines have the highest propensity to generate higher profits and to mitigate their insolvency risk. The non-GCC banks operate genuinely in less developed banking sectors and financial environments compared to their GCC peers. They also operate under stringent banking systems that impede them from mixing banking and non-banking financial activities. Moreover, non-GCC banks that lack the experience and the managerial expertise needed to enter these new market segments might undertake myopic and/or aggressive diversification decisions without being cognizant of the potential risks involved in doing so leading to an exacerbation of the level of risks incurred by the non-GCC banking firms. Consequently, to reach the best performance levels, non-GCC banks should mainly focus on their core traditional banking activities. These results go against those found by Chiorazzo et al. (2008),lsas et al. (2010) and Meslier et al. (2014) for the set of banks in other regions/countries.

All in all, while noninterest income revenue mix has been proven to be the channel through which investment banking business may be conducted in a sound manner for GCC banking firms, non-GCC banks should steer away from noninterest income diversification to avoid the deterioration of their performance.

6. Robustness checks

In order to confirm our findings, we run several robustness checks that provide reassuring results. Firstly, we opt for the t test of the equality of means as another covariate balancing test, as proposed in the literature of continuous treatments impact. We report the t-statistics of the GPS-unadjusted and GPS-adjusted differences in mean values for each covariate and for the three diversification intervals. The first three columns (see Appendix G, Table G.1) correspond to the t-statistics of the GPS-unadjusted sample. It is obvious that the majority of the t-statistics are higher than 1.96, indicating the presence of imbalance in the covariates' distribution. Throughout the robustness checks carried out and for the three samples, namely the whole MENA sample (see Table G.1), the GCC and non-GCC subsamples (see, respectively, Tables G.2 and G.3 in Appendix G), we notice a huge reduction in mean differences, after the *GPS* adjustment (almost all t-statistics are below the value of 1.96). This finding indicates that the *GPS* adjustment greatly improved the covariate balance, corroborating thereby the conclusions based on Bayes factor.[21]

The second robustness test we perform consists in including additional interaction terms between bank- and macroeconomic/industry-specific variables to find out whether our model is sensitive to such modifications or not. Results between bank- and macroeconomic/industry-specific variables are reported in Table G.4 (see Appendix G). For instance, column 1 of Table G.4 reveals that focusing on the whole sample (MENA), banks that do not efficiently manage their operating costs and operate in a weak macroeconomic environment that do not favor the financial innovation (high inflation) tend to diversify their revenue streams as evidenced from/by the significance of the interaction term. Deep analysis by highlighting each region-based bank aside allows for refining the results of the robustness analysis.

When considering the GCC-based banks (column 2 of Table G.4), we notice that banks that enjoy a profitable intermediation activity and operate in an environment that allows them to combine banking and non-banking financial activities have more incentives to diversify and to widen their profit margins by engaging in multiple businesses. Also, GCC banks that are rapidly growing their assets and operate in a sector where banking activity restrictions are in their lowest levels (a financial sector that allows them to expand) are more prone to shift towards off-balance sheet activities. In the same vein, results from non-GCC-based banks (column 3 of Table G.4 in Appendix G) demonstrate that the inefficient cost management attitude prompts non-GCC banking firms to enter new market segments that generate noninterest income when there is a certain freedom in banking activity. Moreover, operating in a relatively developed financial market gives non-GCC banks with high credit risk exposure more incentives to diversify towards off-balance sheet activities. Overall, our results remain unchanged even when adding interaction terms. Panels A, B and C of Table G.5 correspond to the estimated DRFs for the whole MENA, GCC, and non-GCC samples, respectively.

As a third robustness check, we verify whether our baseline results hold when the crisis dummy is considered. For this purpose, we run three regressions, one for each panel country, and depict the results in columns 1, 2 and 3 of Table G.6 (see Appendix G). When spotlighting MENA-based banks' case, results indicate that banks with a rapid asset expansion tend to shift away from noninterest income diversification during the GFC period (see column 1 of Table G.6). Likewise, when GCC-based banks are concerned (see Table G.6, column 2), we notice that banks that are profitable in their traditional business lines are steering away from diversification during the financial turmoil. Furthermore, GCC banking firms with low liquidity risk tend to diversify their revenue streams during the GFC period. A low level of liquidity risk means that GCC banks are less vulnerable to failure and thus, they may diversify their income sources. Finally, when emphasizing non-GCC banks (see Table G.6, column 3), regression results show that banks with rapid growth in

[21] For space constraints, we only report the tables of the *GPS* estimation. The balancing property tests as well as the *DRF* graphs are available from the authors upon request.

their assets seem reluctant to diversify during the GFC. Hence, our results remain almost unchanged. Panels A, B and C of Table G.7 refer to the estimated DRFs for the whole MENA, GCC, and non-GCC samples, respectively.

As a fourth robustness test, we check whether banks' behavior changes in the aftermath of the GFC. As with the previous robustness checks, we run three regressions for the three bank samples (i.e., MENA, GCC and non-GCC). Results are reported in Table G.8 (see Appendix G). It stands out from these results that when it comes to MENA banks (column 1 of Table G.8), results show that banks prefer focusing on their traditional activities in the postcrisis era. In addition, MENA banking firms that do not efficiently manage their operating costs are prone to diversify in the aftermath of the GFC. Results from the GCC sample (column 2 of Table G.8) reveal that banks that are profitable in their lending business lines tend to prefer the diversification shifts, after the financial turmoil era (the attitude reversed in the aftermath of the GFC). Finally, for non-GCC (column 3 of Table G.8), the attitude of those with rapid asset expansion tends to be the same as in the crisis period (stays unaltered). Put simply, these banking firms prefer steering away from nontraditional activities. Panels A, B, and C of Table G.9 correspond to the estimated DRFs for the whole MENA, GCC, and non-GCC samples, respectively.

We perform a fifth robustness check consisting in analyzing banks' behavior before the GFC (see Table G.10 in Appendix G). It stands out from the estimates that before the GFC, MENA-based banks with high equity capital are shifting away from revenue diversification. Again, MENA banking firms that are rapidly growing their assets diversify their revenue streams in the precrisis period (see column 1 of Table G.10). By the same token, when focusing on GCC-based banks (column 2 of Table G.10), estimates reveal that liquid banks tend to shift away from diversification during the precrisis era. However, when it comes to non-GCC based banks (column 3 of Table G.10), results indicate that banks with rapid asset expansion have high propensity to enter new market segments that generate noninterest revenue before the GFC. Panels A, B and C of Table G.11 refer to the estimated DRFs for the whole MENA, GCC and, non-GCC samples, respectively.

In addition to the global crisis, we run a sixth robustness check regarding political instability during the "Arab Spring" crisis. For this purpose, we attempted to consider the 2011 political crisis by creating a dummy variable –impact11- taking the value of 1 if the year into consideration is 2011 and 0 otherwise. We rerun the GPS-DRF model for the MENA, CGG and non-GCC subsamples (see Table G.12 in Appendix G). For the MENA sample, the coefficient of impact11 was statistically insignificant. Hence, the findings indicate no apparent/clear-cut direct impact of the 2011 political turmoil on MENA banks' decision to diversify their revenue streams. As expected, the impact was statistically insignificant for the GCC subsample of banks even though the coefficient of the variable "impact11" exhibited a positive sign (see Table G.12, column 2). As for the non-GCC based banks, the impact of the 2011 political crisis dummy became statistically significant; its coefficient bears a negative sign (see Table G.12, column 3). As expected, the so-called "Arab Spring" adversely affected non-GCC banks' propensity to shift toward nontraditional activities. The political instability that the region witnessed resulted in fragile and less efficient macroeconomic environments, a substantial decrease in financial market efficiency, and low prospects for financial development and thus gave non-GCC banks more incentives to shift away from nontraditional activities that might exacerbate not only their risk profile or mitigate their profitability but also adversely impact the overall banking sector sustainability.

Once again, we run the balancing property test using the Bayes factor for the GPS estimation of the non-GCC subsample. The results continue to indicate that the balancing property is satisfied. We estimate the conditional expectation(s) of the outcome(s) raroaa, raroae, and z-score (see Table G.13 in Appendix G). The resulting DRFs associated with raroaa, raroae, and z-score are displayed in Appendix H. The figures exhibit similar shapes to those of the non-GCC-based banks of the baseline model. Thus, considering the so-called "Arab Spring" does not change the fact that non-GCC banks need to focus on specific nontraditional products that are not market-based (since the latter are volatile and depend on market fluctuations). Non-GCC banks operate under unfavorable conditions due to the 2011 political turmoil (fragile political atmosphere and unfavorable macroeconomic and financial conditions).

We perform a seventh robustness check to ensure that the inference is not surrounded by the selection bias induced by the distribution of banks.[22] As Bahrain and Lebanon have the highest number of banks in their respective subsamples (GCC and non-GCC subsamples) compared to the rest of the MENA countries, we assess hereinafter the likely impact of the sample composition on the GPS estimates. We check how the sample composition affects our empirical findings by re-estimating the model for "non-Bahrain" and "non-Lebanon" subsamples. The estimates of the GPS for the GCC subsample without Bahrain are reported in Appendix G (Table G.14, Panel A) whereas those of non-GCC subsample without Lebanon are reported in Appendix G (Table G.14, Panel B). To evaluate the quality of the GPS estimation, we run the balancing property check via the Bayes factor. The latter clearly shows that the balancing property is satisfied.[23] Once again, we estimate the conditional expectation(s) of the outcome(s) raroaa, raroae, and z-score (see Table G.15 in Appendix G). The estimated DRFs are depicted in Appendix I (Figs. 1–3) and Appendix J (Figs. 1–3) for the GCC and non-GCC subsamples, respectively. The figures exhibit similar shapes to those of the baseline model. Thus, after discarding Bahrain and Lebanon, the DRFs associated with the GCC subsample continue to display an inverse U-shaped curve while its peers related to the non-GCC subsample of banks still exhibit a U-shaped curve. This robustness test corroborates the results of the baseline model in that GCC banks reap benefits in the form of higher performance when they diversify their noninterest income streams. It is preferable for their non-GCC counterparts to focus on specific nontraditional business lines. In a nutshell, our findings appear to be robust and thus, are not driven by the presence of a high number of Bahraini and Lebanese banks in the GCC and non-GCC subsamples, respectively. Thus, sample concentration has no impact on our estimation results.

[22] The authors are grateful to an anonymous reviewer who suggested such a robustness check.

[23] For brevity purposes, we do not report the detailed results with regard to the Bayes factor test but they can be obtained from the authors upon request.

To vindicate that the results are not affected by bank type,[24] we perform an eighth and last robustness check. To account for bank type, we constructed three dummy variables that capture the main bank types (in our MENA sample), spec_C, spec_ISL and spec_INV, standing for commercial, Islamic and investment banks, respectively. Results show that the coefficient of the commercial bank dummy (spec_C) is negative and statistically significant for MENA and non-GCC samples (see Panels A and B, respectively, in Table G.16, Appendix G). These findings can be explained by the fact that MENA commercial banks are focusing on developing commission- and fee-based products (specific types of nontraditional activities) and shifting away from trading and market-oriented business lines that are generally risky. These banking firms already have strong foothold in the loan and deposit markets and do not perceive any distinct gain from engaging in new market segments in which they have no/or little experience and expertise.

Likewise, the results (see Panel A of Table G.17 in Appendix G) indicate that the coefficient estimate of the dummy covariate (spec_INV) is statistically significant and bears a negative sign. These findings reveal that MENA investment banks appear less inclined to diversify noninterest income sources when compared with other non investment banks. Although these banking firms generate a large share of their income from nontraditional activities such as trading and investment banking, they prefer focusing on prespecified source of noninterest income (e.g., fees and commissions) and try to mitigate their exposition to market fluctuations by limiting their market-oriented activities (as the latter are risky), especially after the political instability that the MENA region witnessed and that resulted in a substantial decrease in financial market efficiency.[25]

Lastly, as IBs dominated in GCC banking system, we estimated the GPS for the GCC subsample while differentiating between Islamic and non-IBs through the inclusion of the dummy variable, spec_ISL. Results (see Panel B, Table G.17 in Appendix G) appear very interesting insofar as the coefficient of spec_ISL dummy variable is positive and statistically significant. Such a result indicates that IBs have more incentives to expand toward nonfinancing activities to reach higher profitability and more financial stability. To assess the quality of the GPS estimation after taking into account bank specialization, we evaluate the extent to which the balancing property was verified by having recourse to the Bayes factor. The test reveals that conditioning on the GPS has balanced the measured pre-diversification covariates between diversified banking firms and their concentrated counterparts. Then, we estimate the DRF. Table G.18 (Appendix G) reports the estimation results of the conditional expectation(s) of the outcome(s) raroaa, raroae, and z-score. The resulting DRFs related to raroaa, raroae, and z-score for the MENA, GCC and, non-GCC samples are depicted in Appendix K, Appendix L and Appendix M, respectively.

Even after accounting for bank type (and thus, for bank business model), the DRFs associated with MENA and non-GCC samples still exhibit a U-shaped curve for the three performance indicators. At the same time, they display an inverse U-shaped curve for their GCC peers. The obtained results continue to indicate that Islamic GCC banks that expand towards new market segments and that generate nonfinancing revenue reap benefits from higher profitability and better financial stability. The investment and commercial MENA and non-GCC banking firms need to focus on specific types of noninterest income-generating business lines to reach the best performance levels.

In sum, the above results clearly show that the DRFs of the benchmark/baseline models are robust to the different tests, the different model specifications and the subsample analysis carried out.

7. Concluding remarks and policy implications

The main objective of this study has been to analyze the implications of the trend of shifting toward noninterest income-generating activities in the MENA banking sector in terms of risk-adjusted profitability and insolvency risk. To do so and to account for potential selection bias, we have employed the GPS method that has allowed for continuous treatments, different levels of income mix. Then, we estimated a DRF to depict the expected MENA banks' performance, given the level of diversification adopted and the estimated GPS. This paper has provided strong evidence that expanding into nontraditional activities adversely affects MENA banks' profitability and decreases their financial stability.

Sample splits have showed disparities in terms of the noninterest income mix-bank performance nexus. For the GCC banks, revenue diversification has had the potential to improve their future profits and to mitigate risks. In contrast, less revenue mix, lower noninterest revenue levels and more concentration toward mainstream traditional activities have slowly decreased their performance. However, as non-GCC banks shift toward noninterest income, their risk-adjusted profitability and financial stability have gradually deteriorated until reaching their lowest levels at the noninterest income dose, at which the perfect mix has been reached. The performance resulting from the engagement in multiple nontraditional activities has been lower than the case that those same non-GCC banks have opted for less noninterest income mix and specialized in specific individual activities.

The findings should be of a paramount interest to shareholders, managers, supervisors and policy-makers. For the GCC banks, managers should preponderantly diversify their institutions' revenue streams. Their investments in technology and expertise, coupled with an environment that favors product expansion (developed banking system with a high degree of banking freedom in general), allow for successful shifts toward new market segments, and so lead to higher profitability and more financial sustainability. The GCC banks have more incentives to shift toward off-balance sheet activities such as trading-related business lines (e.g., trading of

[24] We are indebted to an anonymous reviewer who drew our attention to this point.

[25] To corroborate this idea, interacting both the investment bank dummy and the impact11 (of the 2011 political turmoil) in the GPS estimation gave a negative and statistically significant coefficient. This means that investment banks tend to focus on a specific type of noninterest income and limit their exposition to market-based activities due to the fragile political atmosphere and to the unfavorable macroeconomic and financial conditions.

government securities, derivatives, and private securities) to exploit their existing capabilities in order to reap diversification benefits.

However, diversification should be taken with a grain of salt for non-GCC banks as a lower income mix produces the best performance. Non-GCC banks' managers and shareholders tend to overestimate the diversification benefits and underestimate the risks involved in opting for a higher noninterest income mix. Given the macroeconomic conditions in which non-GCC banking firms operate, managers and shareholders should avoid "the overdiversification tendency" that may lead to inefficient and risky business combinations in banking. Thus, managers may opt for low mix of noninterest income, particularly stemming from mainstream intermediation activities (e.g., checking, cash management, letters of credit), and focus on traditional business lines to reach the highest profits and the lowest levels of risk.

Regulatory guidelines and principles should promote noninterest income mix for the GCC banks but not to their non-GCC peers. Regulatory bodies should allocate supervisory resources to GCC banks that enjoy nonrisky mainstream traditional activities to move progressively toward nontraditional activities. Nevertheless, supervision should suffice to assess and monitor the risk profile of each MENA bank. Regulators should supervise large non-GCC banks especially those that do not effectively manage their operating costs and do not take steps to diversify, since a perfect mix of noninterest income engenders a materially higher risk and lower profitability. Close scrutiny should be put on non-GCC banks since diversification has a real side impact on those banking firms that diversify under unfavorable macroeconomic conditions.

Legal barriers to the integration of various financial services and banking consolidation impediments should be dismantled to allow GCC banking firms to enter new market segments and reap diversification benefits in terms of increased performance. The GCC banks with risky loan portfolios have to think twice before undertaking the diversification decision. As these new activities entail potential risks, adding them to the existing ones may exacerbate the risk profile of these banks. A more transparent, complete, and coherent disclosure of the different revenue streams leads to a better understanding of the risks entailed by each diversified GCC bank.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Variables used

Table A.1
Variable definitions.

Variable	Proxy	Definition
		Panel A: Treatment (diversification) measure
div_non	Herfindahl Hirschmann concentration Index	$HHI = \left(\frac{COM}{NON}\right)^2 + \left(\frac{TRAD}{NON}\right)^2 + \left(\frac{OTHOP}{NON}\right)^2$ where COM, TRAD, OTHOP, and NON stand for commission and fee revenues, the trading income, other noninterest revenues and net noninterest income, respectively. Higher values indicate a less noninterest income mix, higher concentration and lower noninterest income share.
		Panel B: Dependent variables
RAROAA	Risk-adjusted return on average assets	Return on average assets over standard deviation of return assets
RAROAE	Risk-adjusted return on average equity	Return on average equity over standard deviation of return equity
$Z-SCORE$	Insolvency risk/ financial stability	Average return on assets and equity to assets to the standard deviation of return assets. It measures insolvency risk
		Panel C: Bank-specific variables
size	Bank size	Natural logarithm of bank total assets
capitalratio	Capital ratio	Book value of equity to total assets
liquidity	Liquidity	Liquid assets to total assets
llp_r	Loan loss provision ratio	Loan loss provision to total assets
dta	Deposits ratio	Deposits to total assets
nim	Net interest margin ratio	Interest income relative to earning assets
bank_growth	Growth rate of bank assets	Annual growth rate of real bank assets
bankefficiency	Bank efficiency	Ratio of earning assets to total bank assets
		Panel D: Industry-specific variables
gdp_growth	External macroeconomic determinant	Annual real gross domestic product growth
inf	External macroeconomic indicator	Annual consumer price inflation
banking_fr	Financial freedom	Country-specific annual index of financial freedom
findev	Financial development	Annual market capitalization to gross domestic product

Appendix B. Descriptive statistics

Table B.1
(Descriptive statistics GCC, non-GCC, and whole sample).

Covariate	Obs	Mean	Std. Dev.	Min	Max
			GCC subsample		
div_non	1469	0.534	0.126	0.333	0.950
raroaa	1330	2.064	1.914	−3.621	10.727
raroae	1330	2.001	1.993	−3.766	12.153
z-score	1330	20.046	20.851	3.049	313.877
size	1469	8.551	2.034	3.049	14.509
capitalratio(%)	1937	25.624	22.316	−15.686	99.326
liquidity (%)	1859	22.041	17.006	0.049	92.727
dta (%)	1468	64.749	23.927	0.021	99.940
llp_r (%)	1365	0.026	0.152	−0.054	1.86
bank_growth(%)	1261	17.865	35.517	−71.73	618.95
gdp_growth(%)	1442	4.810	4.961	−7.076	33.991
bankingfr	1444	68.426	4.996	58.1	77.7
			Non-GCC subsample		
div_non	1694	0.546	0.146	0.333	0.949
raroaa	1665	2.554	3.562	−3.608	42.002
raroae	1665	2.435	2.971	−3.526	24.528
z-score	1665	35.014	64.535	−17.887	829.904
size	1767	7.888	2.049	1.444	15.732
SIZE_SQ	1694	68.037	37.061	2.085	288.783
capitalratio(%)	1694	11.141	8.923	−42.762	86.124
liquidity(%)	1694	31.602	18.576	0.016	93.212
dta (%)	1694	80.317	13.588	0.798	97.586
nim (%)	1694	3.238	1.952	−2.98	24
bank_efficiency	1767	5.837	3.048	0.154	26.719
llp_r (%)	1665	0.605	1.022	−4.772	9.874
bank_growth(%)	1527	16.355	26.988	−55.1	354.06
gdp_growth(%)	1656	4.307	3.348	−13.452	18.665
bankingfr	1581	58.432	5.789	25.67	70.4
findev (%)	951	0.477	0.537	0.037	2.296
			Whole MENA sample		
div_non	2924	0.540	0.139	0.333	0.950
raroaa	2924	2.356	3.002	−3.621	42.002
raroae	2924	2.267	2.633	−3.766	24.528
z-score	2924	28.725	51.213	−17.887	829.904
size	2924	8.153	2.059	1.444	16.951
capitalratio(%)	2693	16.131	17.782	−42.762	99.778
liquidity(%)	2924	27.641	18.218	0.016	94.039
dta (%)	2924	74.158	19.686	0.021	99.940
bank_efficiency	2499	0.981	1.596	0	27.039
llp_r (%)	2924	0.328	0.750	−4.772	9.874
bank_growth(%)	2682	16.984	31.143	−71.73	618.95
inf (%)	2924	4.815	4.603	−4.9	44.964
bankingfr	2798	62.799	7.390	25.67	77.7

Appendix C. Dose–Response Function—GCC sample case

.

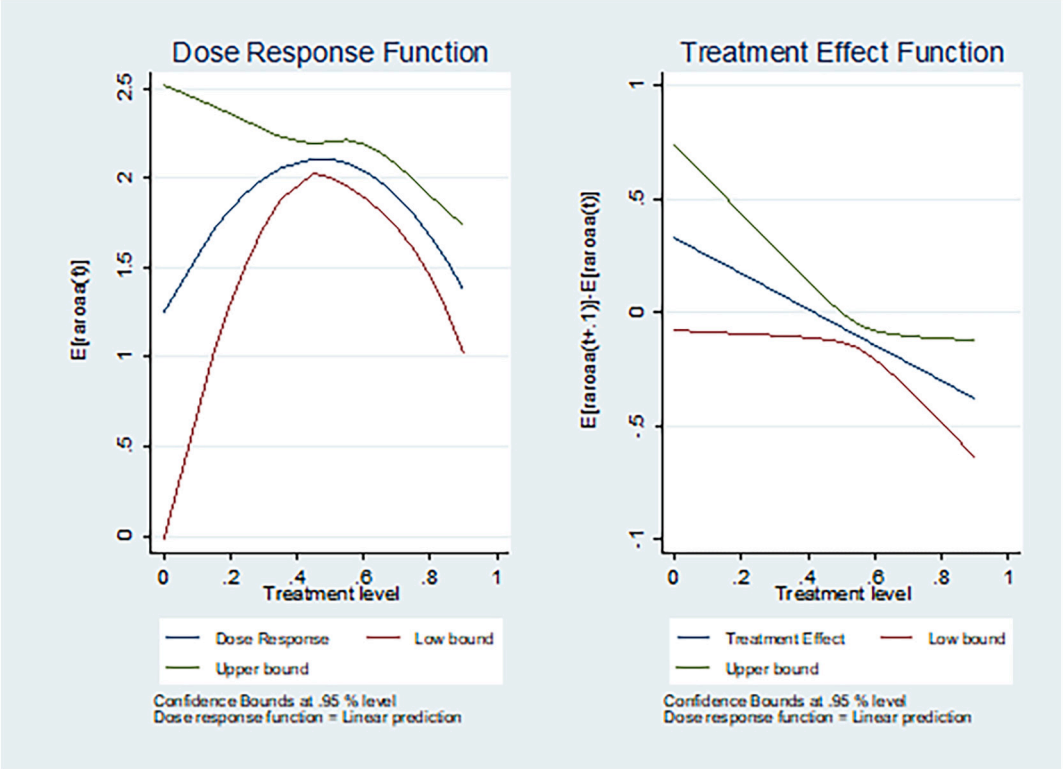


Fig. C.1. DRF and derivative DRF—raroaa.

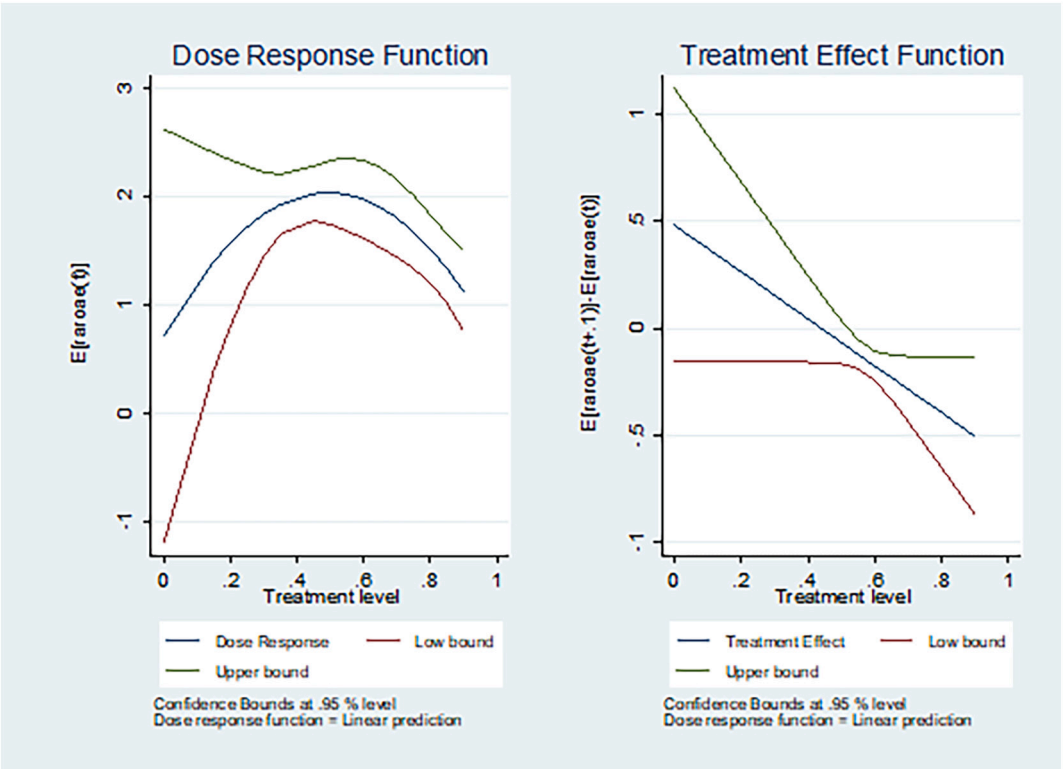


Fig. C.2. DRF and derivative DRF—raroae.

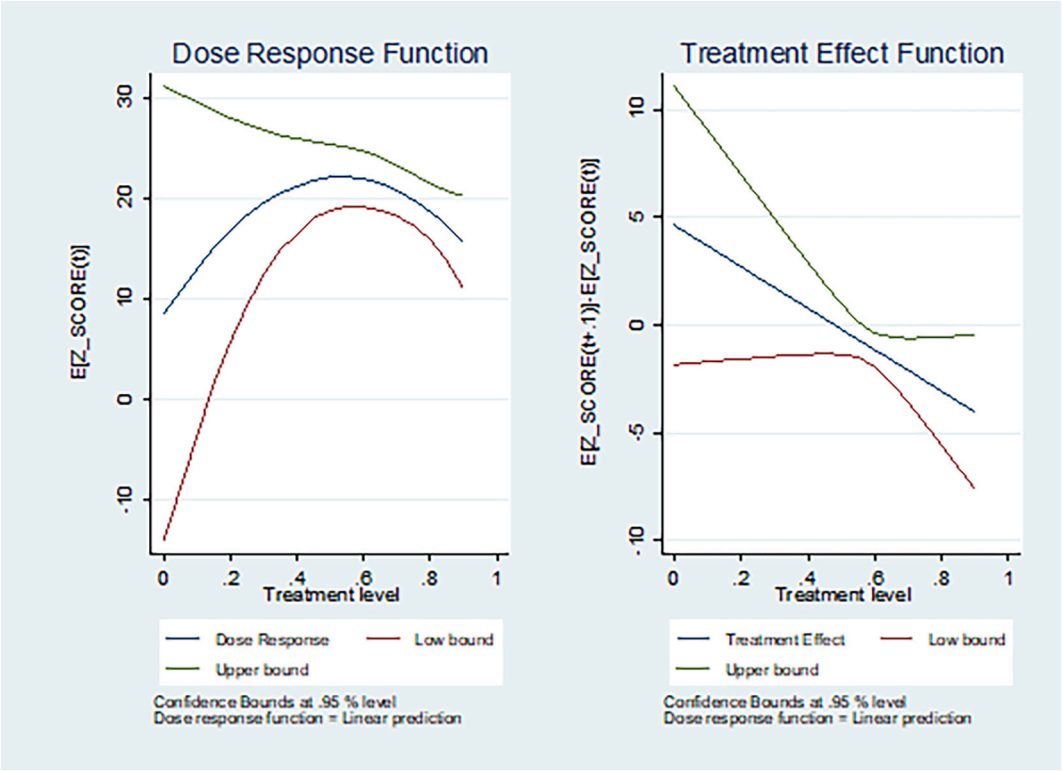


Fig. C.3. DRF and derivative DRF—z-score.

Appendix D. Dose–Response Function—non-GCC sample case

.

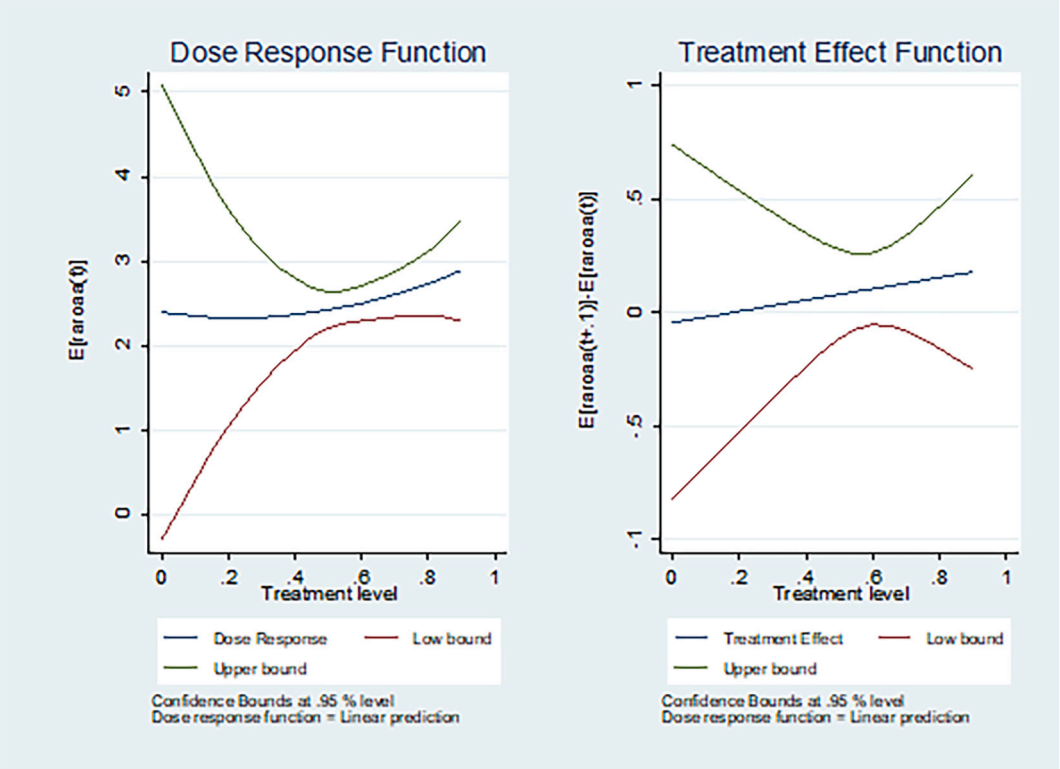


Fig. D.1. DRF and derivative DRF—raroaa.

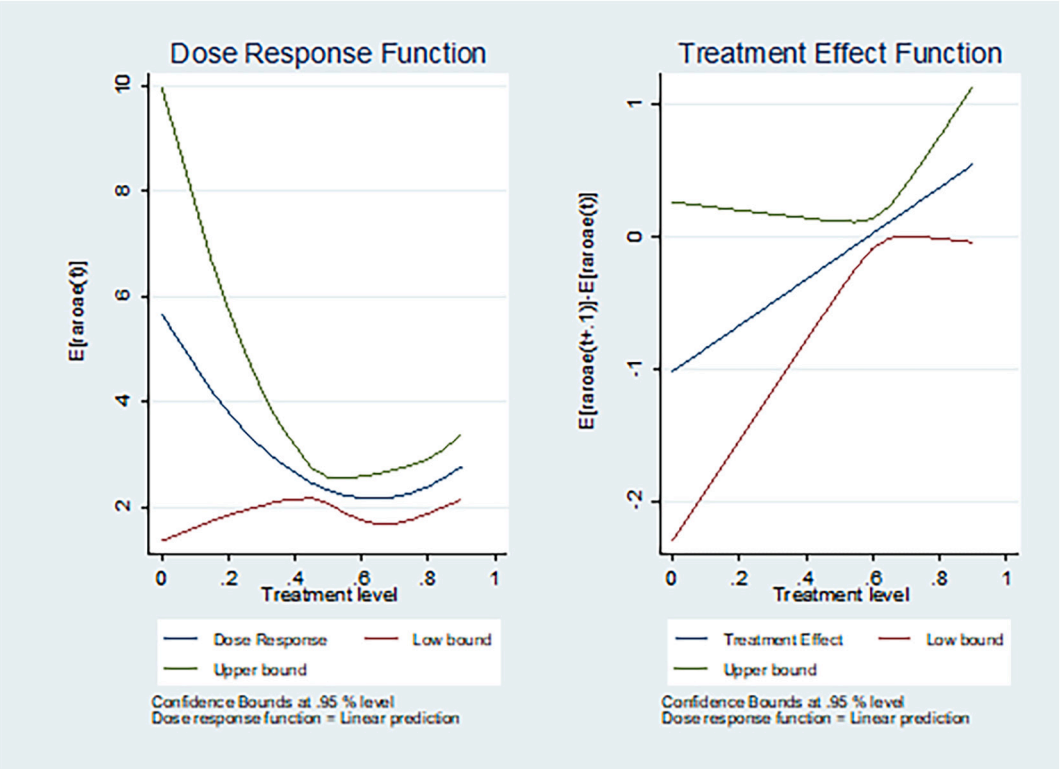


Fig. D.2. DRF and derivative DRF—raroae.

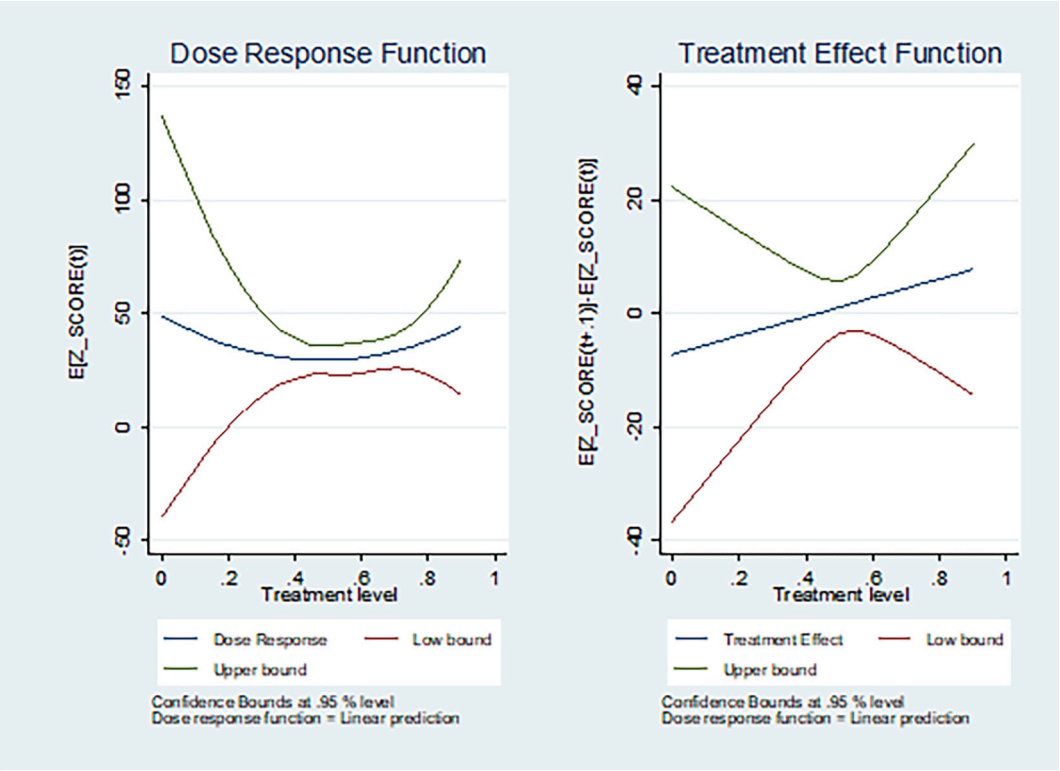


Fig. D.3. DRF and derivative DRF—z-score.

Appendix E. Dose–Response Function—whole MENA sample

.

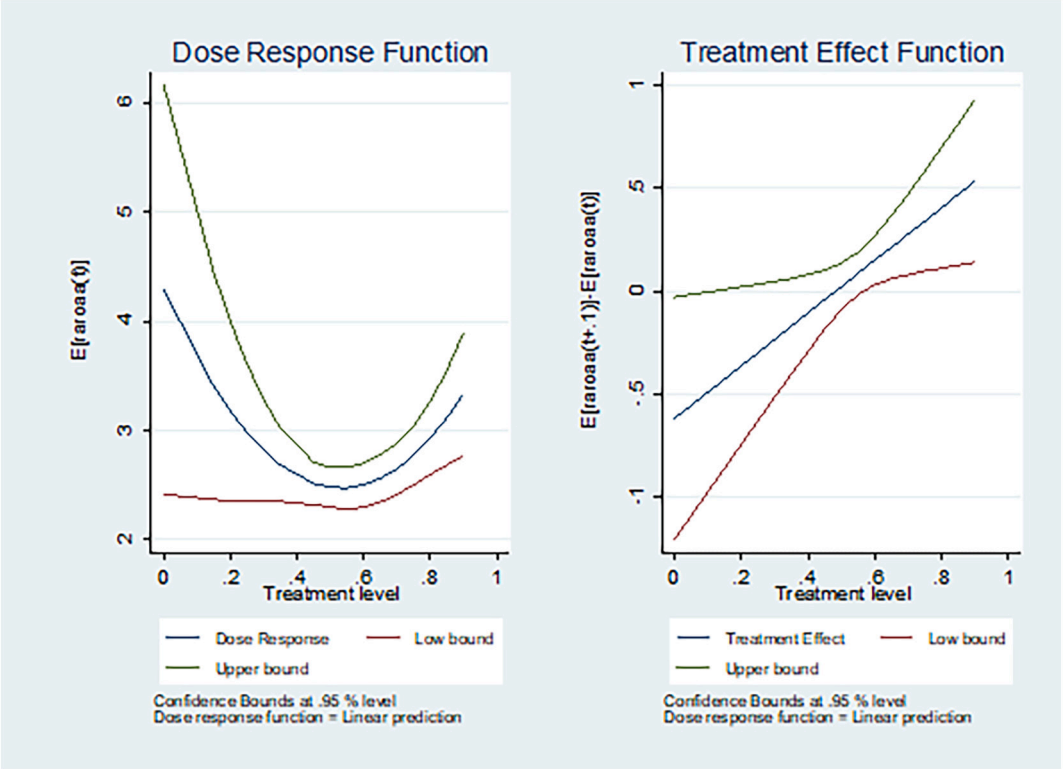


Fig. E.1. DRF and derivative DRF—raroaa.

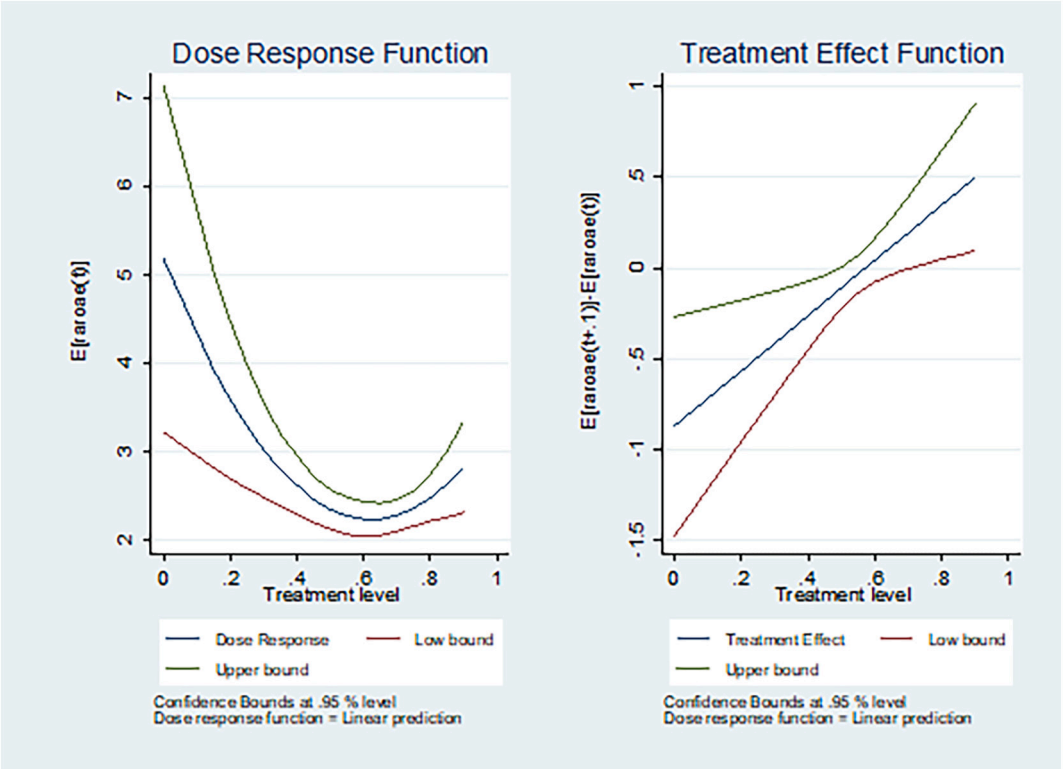


Fig. E.2. DRF and derivative DRF—raroae.

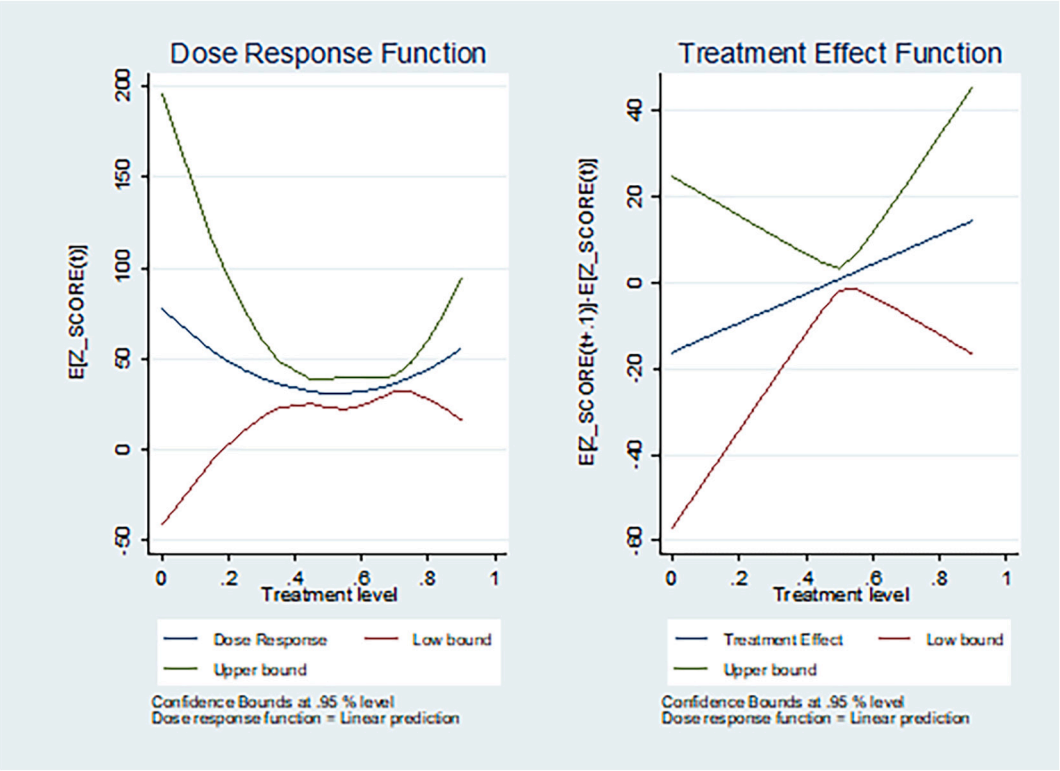


Fig. E.3. DRF and derivative DRF—z-score.

Appendix F. Probability distribution of treatment D (diversification index)

.

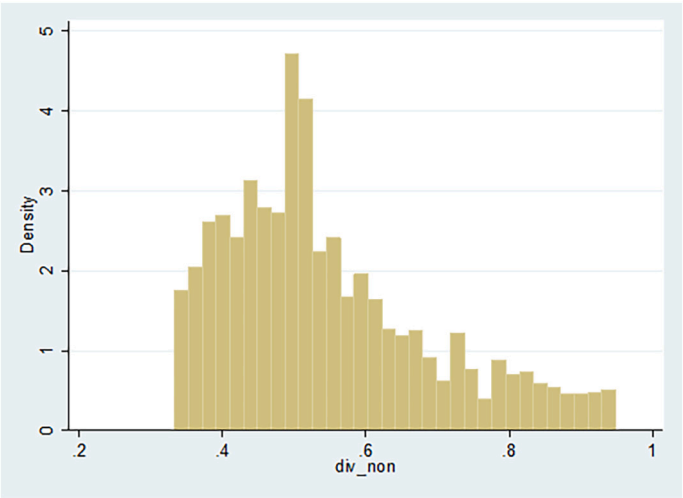


Fig. F.1. Distribution of D—non-GCC sample case.

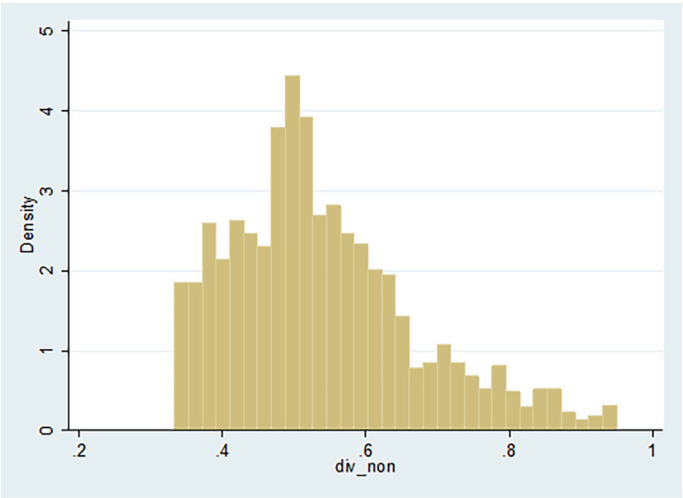


Fig. F.2. Distribution of D—GCC sample case.

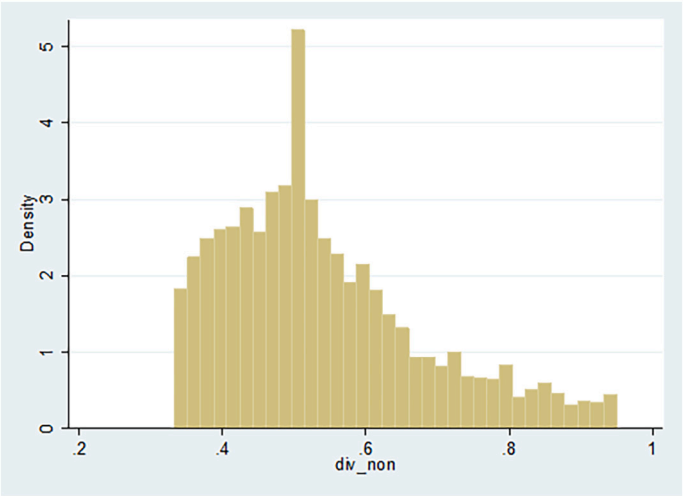


Fig. F.3. Distribution of D—whole MENA sample.

Appendix G. Robustness analysis

Table G.1
Covariate balancing property check (t-test for equality of means—whole sample).

Covariates	GPS-unadjusted			GPS-adjusted		
	TG No. 1	TG No. 2	TG No. 3	TG No 1	TG No 2	TG No 3
size	−3.144	8.036	3.613	−0.892	0.596	−0.666
capitalratio	0.492	−6.001	−3.717	0.208	−0.948	0.542
liquidity	−0.854	−4.230	−2.043	1.358	−1.494	0.389
dta	−0.587	2.723	2.097	0.192	−0.961	0.636
bank_growth	0.979	−1.737	−0.473	−1.727	1.833	0.183
bank_efficiency	1.386	−1.242	0.052	0.687	−1.402	−0.367
llp_r	3.084	−0.220	1.550	−1.826	1.815	−0.543
inf	1.021	1.128	1.771	0.362	−0.652	0.452
findev	−0.795	4.159	2.557	−1.335	−0.356	1.707
bank_efficiency*inf	3.333	0.253	1.884	1.258	−1.772	−0.920

Notes: TG No. i: stands for treatment group No. i (i = 1,…,3). We report the t-values of the standard two-sided t-test.

Table G.2
Covariate balancing property check (t-test for equality of means—GCC subsample).

Covariates	GPS-unadjusted			GPS-adjusted		
	TG No. 1	TG No. 2	TG No. 3	TG No 1	TG No 2	TG No 3
size	−2.714	6.401	2.948	−0.272	−0.831	1.995
capitalratio	1.716	−6.548	−3.440	−0.477	1.380	−1.140
liquidity	−0.760	−4.577	−2.093	1.029	−0.779	0.162
nim	−0.523	−1.102	−1.109	0.574	−0.558	−0.287
dta	−0.695	2.018	1.309	0.719	−1.191	0.911
bank_growth	1.203	−0.553	0.386	−1.482	0.757	0.658
llp_r	−2.312	0.627	−2.046	0.773	−0.867	−0.784
gdp_growth	−0.019	0.842	0.728	1.331	−1.759	1.392
bankingfr	1.594	−1.499	0.102	−0.565	1.478	−1.502
nim*bankingfr	−0.503	−1.266	−1.155	0.550	−0.512	−0.342
bank_growth*bankingfr	1.513	−0.992	0.301	−1.502	0.754	0.627

Notes: TG No. i: stands for treatment group No. i (i = 1,…,3). We report the t-values of the standard two-sided t-test.

Table G.3
Covariate balancing property check (t-test for equality of means— non-GCC subsample).

Covariates	GPS-unadjusted			GPS-adjusted		
	TG No. 1	TG No. 2	TG No. 3	TG No 1	TG No 2	TG No 3
size	−1.583	4.748	2.217	−0.779	0.825	−0.795
SIZE_SQ	−1.355	3.619	1.541	−0.720	0.770	−0.744
capitalratio	−1.117	−5.198	−3.465	0.867	−0.108	−0.746
liquidity	−0.692	−0.495	−0.993	2.218	−1.873	0.494
dta	−0.447	3.923	3.289	1.063	−1.011	0.673
nim	−2.542	−6.506	−5.990	0.323	0.400	−1.065
bank_efficiency	3.372	0.380	2.031	0.458	−0.591	0.592
llp_r	3.010	0.074	1.690	−2.262	1.845	−0.283
bank_growth	0.002	−1.954	−1.068	−0.141	0.002	−0.437
gdp_growth	0.556	−0.177	0.337	−0.751	0.088	1.150
bankingfr	2.931	2.223	4.268	−1.096	1.802	0.675
findev	0.982	2.984	3.267	−0.711	−0.428	2.242
bank_efficiency*bankingfr	3.466	0.371	1.981	0.849	0.253	−0.723
llp_r*findev	2.822	0.160	1.877	−1.928	2.018	−0.533

Notes: TG No. i: stands for treatment group No. i (i = 1,…,3). We report the t-values of the standard two-sided t-test.

Table G.4
Estimation of the GPS (MENA, GCC, and non-GCC samples).

Covariates	Whole sample	GCC subsample	Non-GCC subsample
size	−0.040***	−0.022**	−0.465***
SIZE_SQ			0.024***
capitalratio	0.006***	0.002**	−0.002
liquidity	−0.032	−0.028	−0.111
nim		−0.054***	0.066***
dta	−0.008	0.023	−0.250
bank_growth	−0.000	−0.015*	0.001
bank_efficiency	−0.051		−0.079
llp_r	0.033	0.435**	−0.049
gdp_growth		0.003	−0.012
findev	−0.002***		−0.001**
bankingfr		−0.006	−0.036***
inf	−0.005**		
bank_efficiency*bankingfr			0.027**
bank_growth*bankingfr		0.000*	
bank_efficiency*inf	0.008***		
nim*bankingfr		0.001***	
llp_r*findev			0.001**
Constant	0.625***	0.660**	4.631***

(*continued on next page*)

Table G.4 (*continued*)

Covariates	Whole sample	GCC subsample	Non-GCC subsample
Log Pseudolikelihood	−637.953	−621.239	−375.102
AIC	0.949	0.950	0.958
Observations	1367	1333	814

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.5
Estimates of the conditional expectation(s) of the outcome(s)—estimated DRFs.

Covariates	Panel A: Whole sample outcomes			Panel B: GCC subsample outcomes			Panel C: Non-GCC subsample outcomes		
	raroaa	raroae	z-score	raroaa	raroae	z-score	raroaa	raroae	z-score
div_non	−13.949*	−11.677*	−721.597***	−14.282*	−11.469	−213.608*	−19.050***	−19.030***	−546.537***
div_non_sq	5.923*	7.437***	119.737	−4.983**	−6.216***	−54.003*	3.156	7.574**	50.170
pscore	−90.752***	−52.924**	−2685.16***	−21.411	8.669	769.119	16.441	16.800	−506.016
pscore_sq	56.928**	31.487	1761.314**	−9.410	−32.851	−836.556	−28.506*	−22.708*	44.237
div_non_pscore	13.693	4.401	1066.959**	35.228**	32.537**	500.125**	27.909**	15.638	863.773***
Constant	37.041***	24.826***	1012.872***	15.333	5.491	−163.895	3.260	3.425	320.389***
Observations	1367	1367	1367	1331	1331	1331	814	814	814

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.6
Estimation of the GPS (MENA, GCC, and non-GCC samples).

Covariates	Whole sample	GCC subsample	Non-GCC subsample
size	−0.045***	−0.024**	−0.396***
SIZE_SQ			0.020***
capitalratio	0.005***	0.002**	−0.005
liquidity	−0.044	−0.077	−0.082
dta	−0.044	0.061	−0.042
nim		0.004***	0.074***
bank_growth	0.000	0.000	0.002
bank_efficiency	0.031*		0.719**
llp_r	0.034	0.415**	0.010
gdp_growth		0.003	−0.010
findev	−0.001***		−0.001***
bankingfr		−0.000	−0.019***
inf	−0.000		
Crisis*liquidity		0.065*	
Crisis*bank_growth	−0.002**		−0.010***
Crisis*nim		−0.005***	
Crisis*capitalratio			0.013***
Constant	0.631***	0.267	3.089***
Log Pseudolikelihood	−614.177	−621.419	−366.811
AIC	0.951	0.950	0.961
Observations	1315	1333	795

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.7
Estimates of the conditional expectation(s) of the outcome(s)—estimated DRFs.

Covariates	Panel A: Whole sample outcomes			Panel B: GCC subsample outcomes			Panel C: Non-GCC subsample outcomes		
	raroaa	raroae	z-score	raroaa	raroae	z-score	raroaa	raroae	z-score
div_non	−11.549	−8.318	−623.929**	−13.174	−10.979	−423.756***	−21.211***	−18.894***	−896.886***
div_non_sq	5.589*	7.439***	147.054	−4.265**	−5.600***	−60.720*	0.392	8.403**	75.720
pscore	−99.014***	−65.618**	−240.319	−77.999**	−68.712*	−1319.315**	11.282	12.214	256.142
pscore_sq	66.294*	46.185	−252.880	40.066	34.748	829.306	−30.457*	−19.687	−888.743**
div_non_pscore	10.395	−1.419	844.513*	31.718*	30.386	898.723***	38.039***	14.388	1445.821***
Constant	38.495***	27.237***	284.547	31.657***	27.718***	476.606***	5.684	5.080	193.512

(*continued on next page*)

Table G.7 (*continued*)

Covariates	Panel A: Whole sample outcomes			Panel B: GCC subsample outcomes			Panel C: Non-GCC subsample outcomes		
	raroaa	raroae	z-score	raroaa	raroae	z-score	raroaa	raroae	z-score
Observations	1315	1315	1315	1331	1331	1331	795	795	795

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.8
Estimation of the GPS (MENA, GCC, and non-GCC samples).

Covariates	Whole sample	GCC subsample	Non-GCC subsample
size	−0.047***	−0.023**	−0.349***
SIZE_SQ		0.002**	0.018***
capitalratio	0.005***	−0.039	0.001
liquidity	−0.048	0.039	−0.099
dta	−0.047	−0.001	−0.042
nim		0.000	0.067***
bank_growth	−0.000	0.003	0.002
bank_efficiency	0.028*	0.423**	0.726*
llp_r	0.034	−0.000	0.011
gdp_growth			−0.014**
findev	−0.001***		−0.001***
bankingfr			−0.015**
inf	−0.000		
Post_Crisis	−0.099*		
Post_Crisis*bank_efficiency	0.102**		
Post_Crisis*nim		0.003*	
Post_Crisis*bank_growth			−0.003**
Constant	0.657***	0.286	2.625***
Log Pseudolikelihood	−614.272	−621.640	−621.640
AIC	0.953	0.949	0.949
Observations	1315	1333	1333

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.9
Estimates of the conditional expectation(s) of the outcome(s)—estimated DRFs.

Covariates	Panel A: Whole sample outcomes			Panel B: GCC subsample outcomes			Panel C: Non-GCC subsample outcomes		
	raroaa	raroae	z-score	raroaa	raroae	z-score	raroaa	raroae	z-score
div_non	−12.162	−8.967	−595.328**	−10.685	−10.456	−231.556	−20.375***	−18.511***	−875.446***
div_non_sq	5.921*	7.582***	152.297	−3.968*	−5.367**	−48.543	1.391	8.827**	88.126
pscore	−149.639***	−79.122**	−1889.652	−18.643	−19.317	−2485.782**	21.811	16.534	259.953
pscore_sq	109.412***	57.596*	1200.587	−13.814	−12.126	2136.044*	−36.965*	−22.456	−854.705*
div_non_pscore	10.692	−0.684	782.807	26.640	29.051	522.540*	34.326**	12.787	1383.103***
Constant	53.397***	31.289***	751.826**	15.393	14.776	727.270**	2.222	3.687	184.380
Observations	1315	1315	1315	1331	1331	1331	795	795	795

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.10
Estimation of the GPS (MENA, GCC, and non-GCC samples).

Covariates	Whole sample	GCC subsample	Non-GCC subsample
size	−0.048***	−0.025**	−0.346***
SIZE_SQ			0.018***
capitalratio	0.007***	0.002**	0.002
liquidity	−0.046	−0.017	−0.110
dta	−0.033	0.071	−0.034
nim		0.000	0.068***
bank_growth	−0.003***	0.000	−0.002**
bank_efficiency	0.030*		0.754**

(*continued on next page*)

Table G.10 (*continued*)

Covariates	Whole sample	GCC subsample	Non-GCC subsample
llp_r	0.036	0.425**	0.010
gdp_growth		0.003	−0.013**
findev	−0.001***		−0.001***
bankingfr		−0.000	−0.015**
inf	−0.000		
Pre_Crisis*liq		−0.073**	
Pre_Crisis*bank_growth	0.003***		0.005***
Pre_Crisis*capitalratio	−0.004**		
Constant	0.657***	0.271	2.598***
Log Pseudolikelihood	−613.914	−621.540	−367.163
AIC	0.952	0.949	0.959
Observations	1315	1333	795

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.11
Estimates of the conditional expectation(s) of the outcome(s)—estimated DRFs.

Covariates	Panel A: Whole sample outcomes			Panel B: GCC subsample outcomes			Panel C: Non-GCC subsample outcomes		
	raroaa	raroae	z-score	raroaa	raroae	z-score	raroaa	raroae	z-score
div_non	−13.913*	−10.005	−652.291***	−5.393	−7.486	−27.573	−20.621***	−18.145***	−874.544***
div_non_sq	5.367	7.361***	134.419	−3.322	−4.985**	−28.764	0.925	8.676**	78.869
pscore	−49.022	−29.203	−745.076	−141.478**	−83.628	−8215.90***	17.406	12.086	150.491
pscore_sq	20.156	12.345	158.824	102.703*	49.531	7432.408***	−34.273*	−18.910	−781.277
div_non_pscore	15.176	1.878	920.035**	15.507	22.750	102.942	35.846***	12.592	1403.742***
Constant	24.929***	17.440**	433.712	47.897***	31.618**	2281.539***	3.650	4.965	218.469
Observations	1315	1315	1315	1331	1331	1331	795	795	795

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.12
Estimation of the GPS (MENA, GCC, and non-GCC samples).

Covariates	Whole sample	GCC subsample	Non-GCC subsample
size	−0.039***	−0.033***	−0.482***
SIZE_SQ			0.026***
capitalratio	0.006***	0.002**	−0.002
liquidity	−0.012	−0.050	−0.127
dta	−0.086	0.045	−0.376*
nim		0.001	0.052***
bank_efficiency	−0.001		0.03
llp_r	0.035	1.769	0.011
bank_growth	0.001	0.001	0.001
gdp_growth		0.003	−0.011
bankingfr		0.004	−0.015**
inf	0.002		
findev	−0.002***		−0.001***
impact11	−0.059	0.054	−0.225*
Constant	0.718***	0.050	3.602***
Log Pseudolikelihood	−654.139	−632.739	−413.031
AIC	0.954	0.952	0.956
Observations	1.394	1.352	893

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.13
Estimates of the conditional expectation(s) of the outcome(s)—estimated DRFs.

Covariates	non-GCC subsample

(*continued on next page*)

Table G.13 (*continued*)

Covariates	non-GCC subsample		
	raroaa	raroae	z-score
	raroaa	raroae	z-score
div_non	−16.769**	−15.100***	−406.389*
div_non_sq	3.677	6.897**	151.734
pscore	49.448**	25.092	1788.697***
pscore_sq	−55.184***	−28.665*	−1678.942***
div_non_pscore	22.880*	10.562	422.120
Constant	−6.489	0.324	−384.878**
Observations	888	836	874

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.14
Estimation of the GPS.

Covariates	Panel A: GCC (without Bahrain)	Panel B: Non-GCC (without Lebanon)
size	−0.013	−0.364***
SIZE_SQ		0.020***
capitalratio	0.002*	0.006
liquidity	−0.061*	−0.260**
dta	0.060	−0.205
nim	0.000	0.030*
bank_efficiency		0.542*
llp_r	2.321	0.045
bank_growth	−0.001	−0.001
gdp_growth	0.004	−0.010
bankingfr	−0.001	−0.031***
findev		0.001
Constant	0.187	3.605***
Log Pseudolikelihood	−479.077	−261.993
AIC	0.952	0.975
Observations	1.028	564

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.15
Estimates of the conditional expectation(s) of the outcome(s)—estimated DRFs.

Covariates	Panel A: GCC sample outcomes			Panel B: non-GCC subsample outcomes		
	raroaa	raroae	z-score	raroaa	raroae	z-score
div_non	−5.342	−8.616	−831.292***	−20.279**	−13.469*	−874.002***
div_non_sq	−3.443	−4.536**	−50.427	3.693	7.635*	245.599*
pscore	67.209	48.427	−6083.602***	100.236***	76.672***	1981.351***
pscore_sq	−84.998	−69.373	5149.760***	−95.791***	−65.272***	−2120.007***
div_non_pscore	15.365	23.975	1643.884***	30.941*	6.738	1113.555 **
Constant	−9.855	−4.779	1790.307***	−22.619***	−16.864***	−344.025
Observations	1.028	1.028	1.028	564	564	564

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.16
Estimation of the GPS (accounting for bank type).

Covariates	Panel A: MENA sample	Panel B: non-GCC subsample
size	−0.042***	−0.441***
SIZE_SQ		0.023***
capitalratio	0.005***	−0.002
liquidity	−0.005	−0.084
dta	−0.064	−0.162

(*continued on next page*)

Table G.16 (*continued*)

Covariates	Panel A: MENA sample	Panel B: non-GCC subsample
nim		0.050***
bank_growth	0.000	0.000
bank_efficiency	−0.001	0.588*
llp_r	0.036	0.014
inf	0.001	
gdp_growth		−0.012*
bankingfr		−0.014**
findev	−0.002***	−0.002***
spec_C	−0.117**	−0.152**
Constant	0.839***	3.349***
Log Pseudolikelihood	−653.771	−401.190
AIC	0.954	0.956
Observations	1.394	869

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.17
Estimation of the GPS (accounting for bank type).

Covariates	Panel A: MENA sample	Panel B: GCC subsample
size	−0.042***	−0.032***
capitalratio	0.007***	0.002*
liquidity	−0.014	−0.031
dta	−0.093	0.028
nim		0.001
llp_r	0.035	2.374
bank_growth	0.001	0.001
bank_efficiency	−0.001	
gdp_growth		0.003
bankingfr		0.007*
inf	0.002	
findev	−0.002***	
spec_INV	−0.132**	
spec_ISL		0.081*
Constant	0.746***	−0.108
Log Pseudolikelihood	−653.967	−600.770
AIC	0.954	0.951
Observations	1.394	1.286

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Table G.18
Estimates of the conditional expectation(s) of the outcome(s)—estimated DRFs (accounting for bank type).

Covariates	Panel I: MENA sample			Panel II: GCC subsample			Panel III: Non-GCC subsample		
	raroaa	raroae	z-score	raroaa	raroae	z-score	raroaa	raroae	z-score
iv_non	−9.214	−6.548	−91.544	−12.808	−10.482*	−347.586**	−13.279	−11.460*	−201.863
div_non_sq	−2.373	−3.265	−42.682	3.445	3.424	65.294	3.461	3.329	82.146
pscore	−34.104	41.758	405.214	−80.067**	−31.947	−661.349	−49.158	−23.818	−1074.737
pscore_sq	4.178	−58.304	−464.640	49.322	13.527	206.097	23.323	6.294	718.603
div_non_pscore	19.913	16.895	248.012	16.042	9.709	501.448*	16.669	11.486	210.832
Constant	19.306	−3.802	−71.551	33.037***	17.835**	354.768	23.965**	15.574*	429.006*
Observations	1.286	1.286	1.286	1.394	1.394	1.394	1.394	1.394	1.394

Notes: ***, **, and * indicate the statistical significance at the 1%, 5% and 10%, respectively.

Appendix H. DRFs—Robustness check: Impact of 2011 political turmoil (non-GCC sample)

.

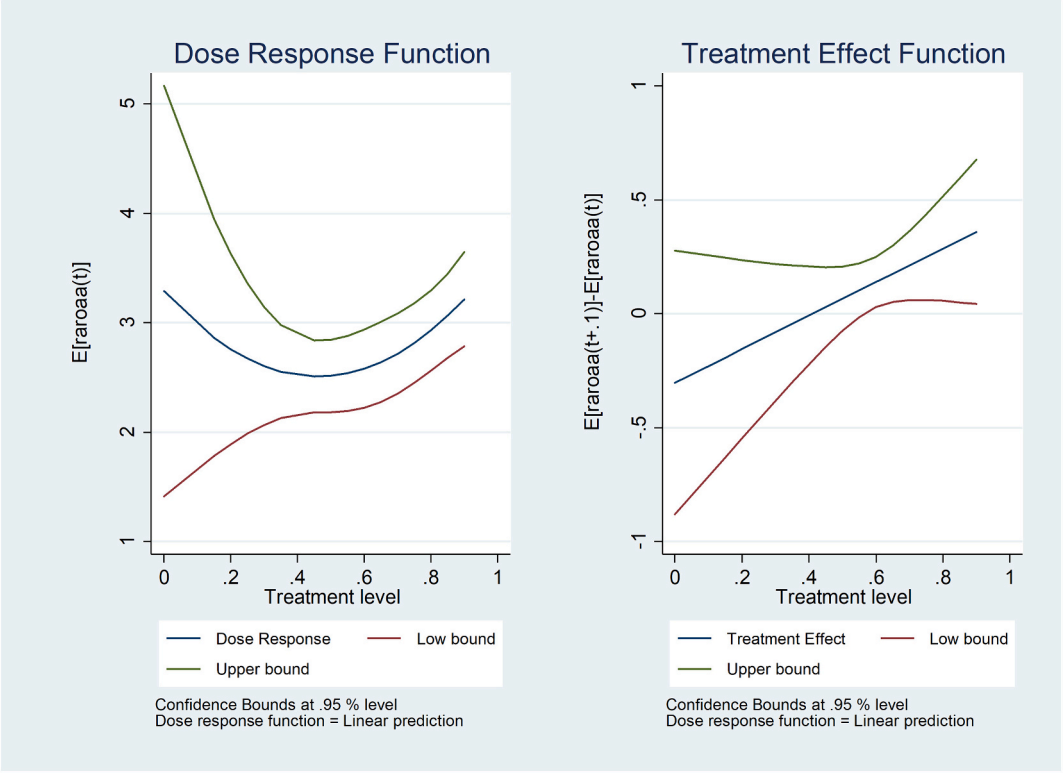


Fig. H.1. DRF and derivative DRF—raroaa.

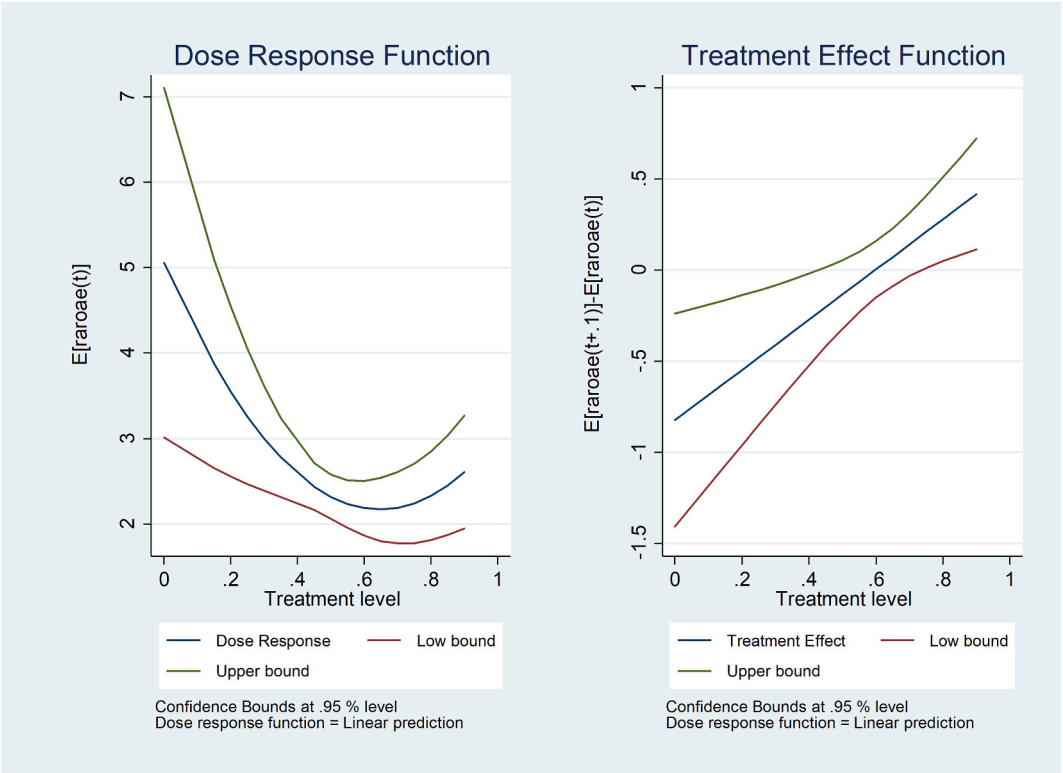


Fig. H.2. DRF and derivative DRF—raroae.

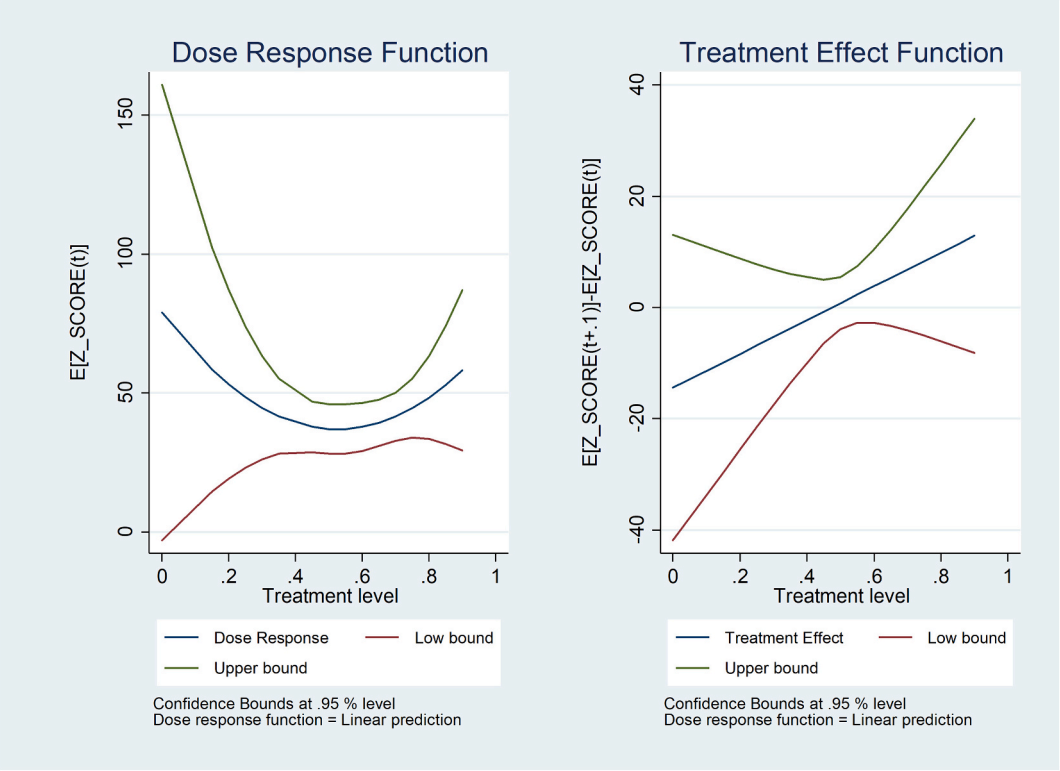


Fig. H.3. DRF and derivative DRF—z-core.

Appendix I. DRFs—Robustness check: Sample selection bias (GCC sample)

.

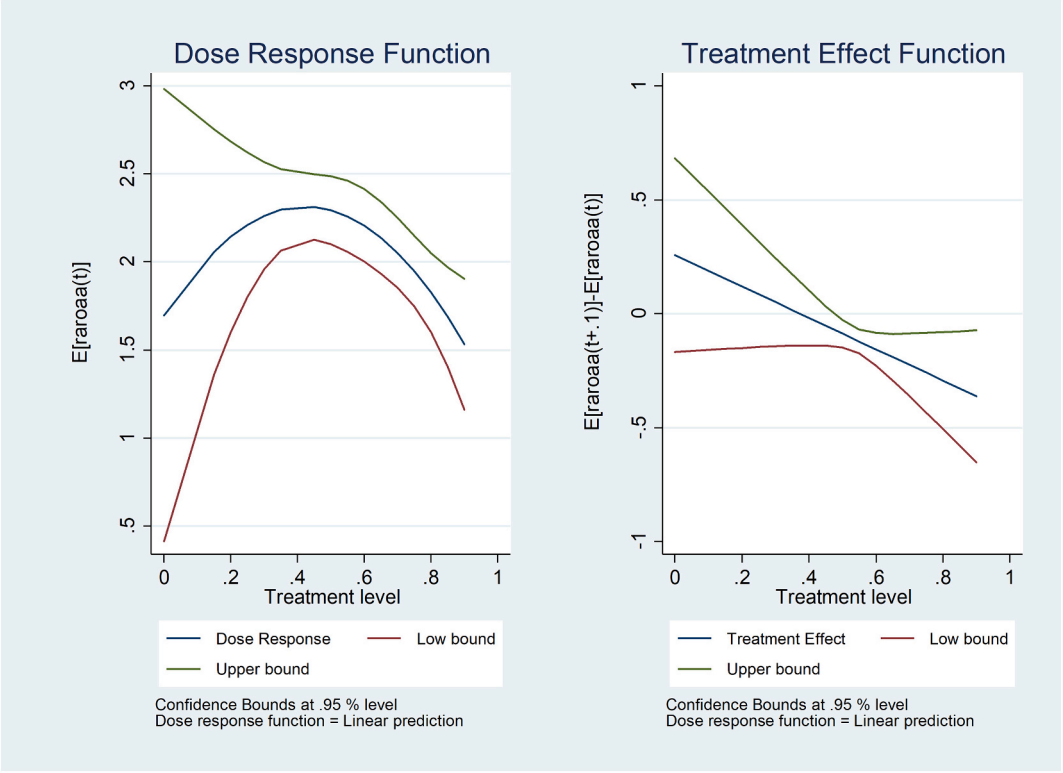


Fig. I.1. DRF and derivative DRF—raroaa.

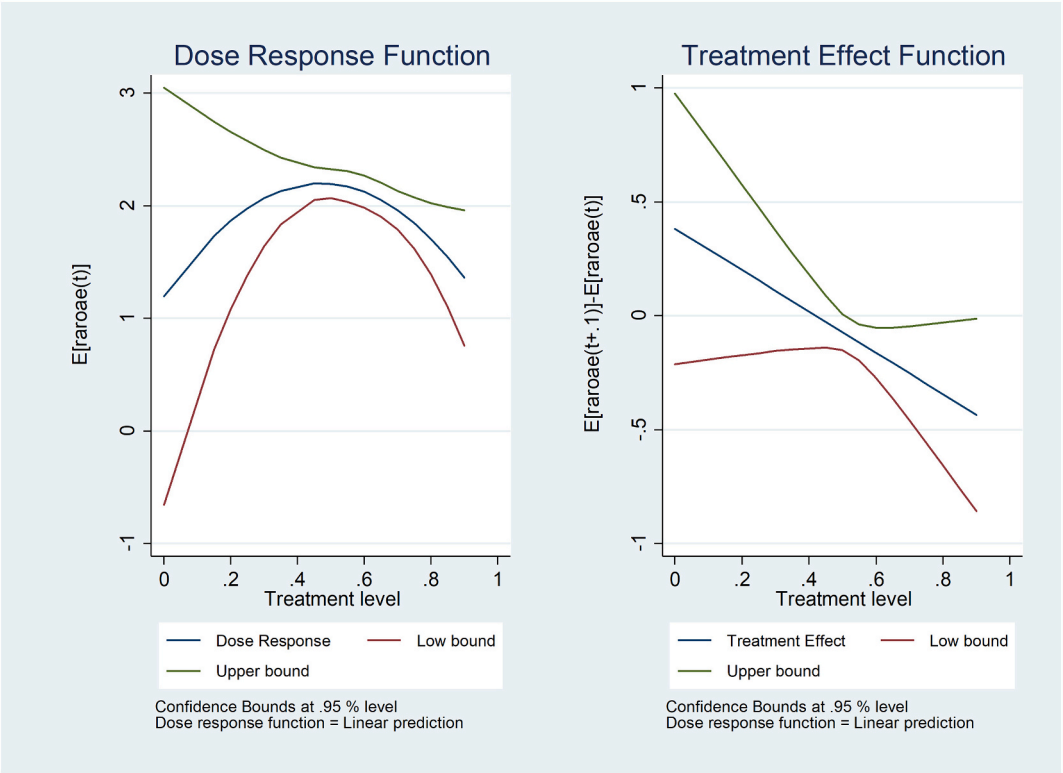


Fig. I.2. DRF and derivative DRF—raroae.

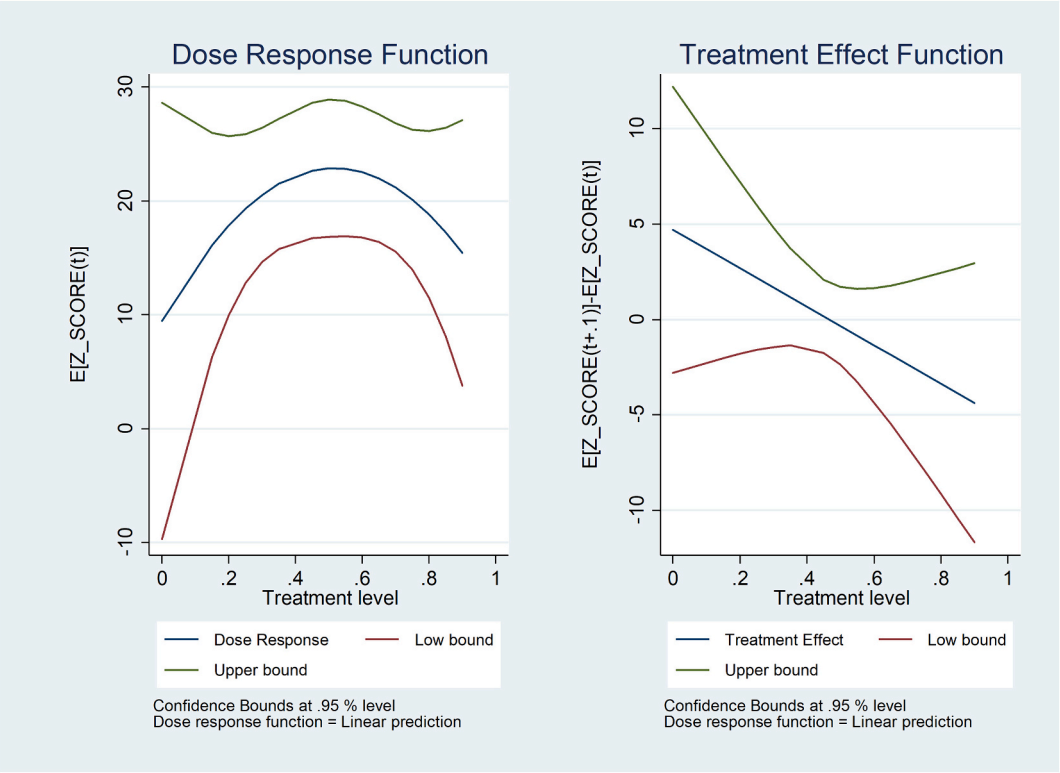


Fig. I.3. DRF and derivative DRF—z-score.

Appendix J. DRFs—Robustness check: Sample selection bias (non-GCC sample)

.

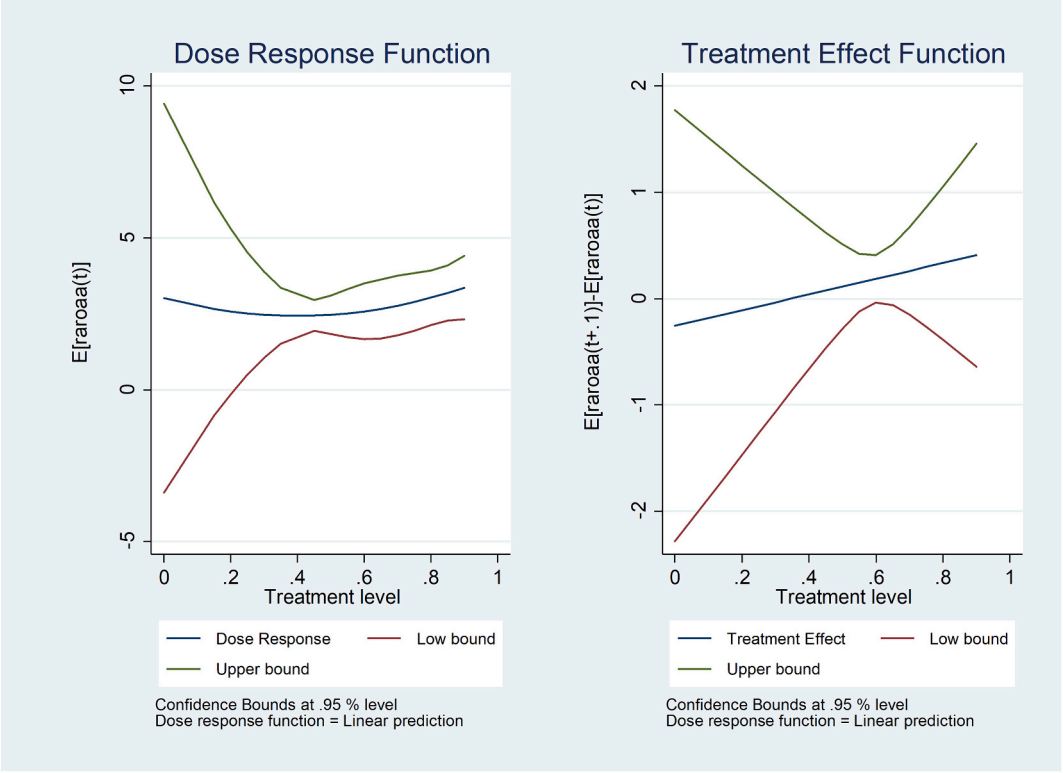


Fig. J.1. DRF and derivative DRF—raroaa.

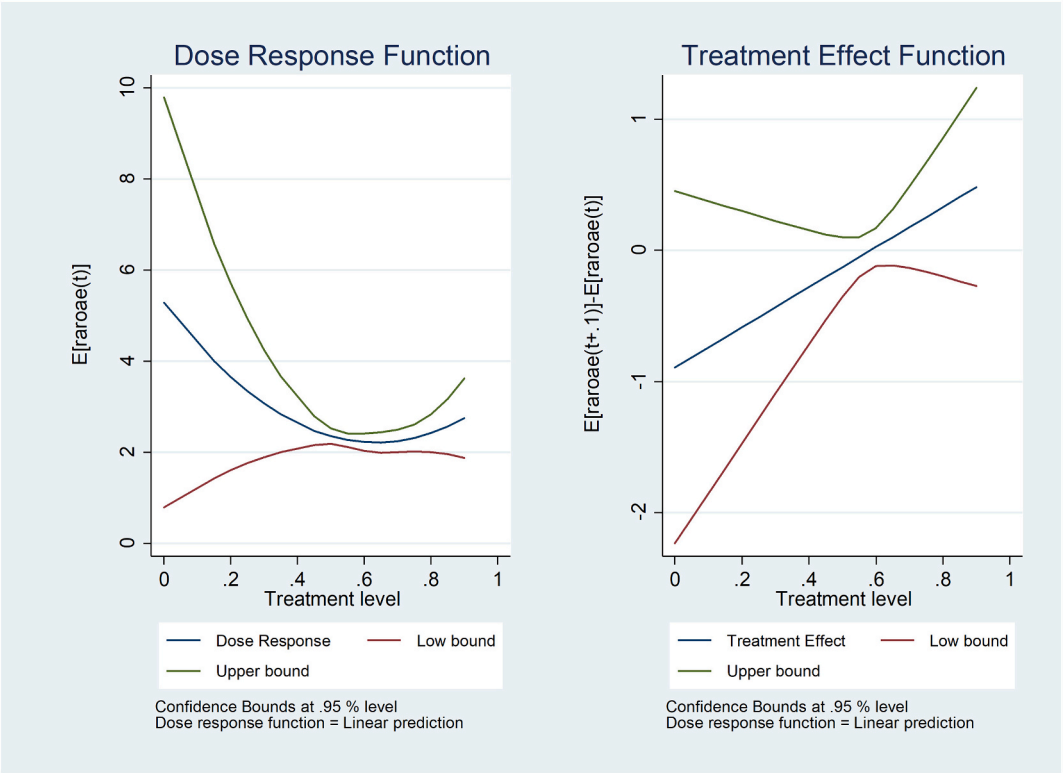


Fig. J.2. DRF and derivative DRF—raroae.

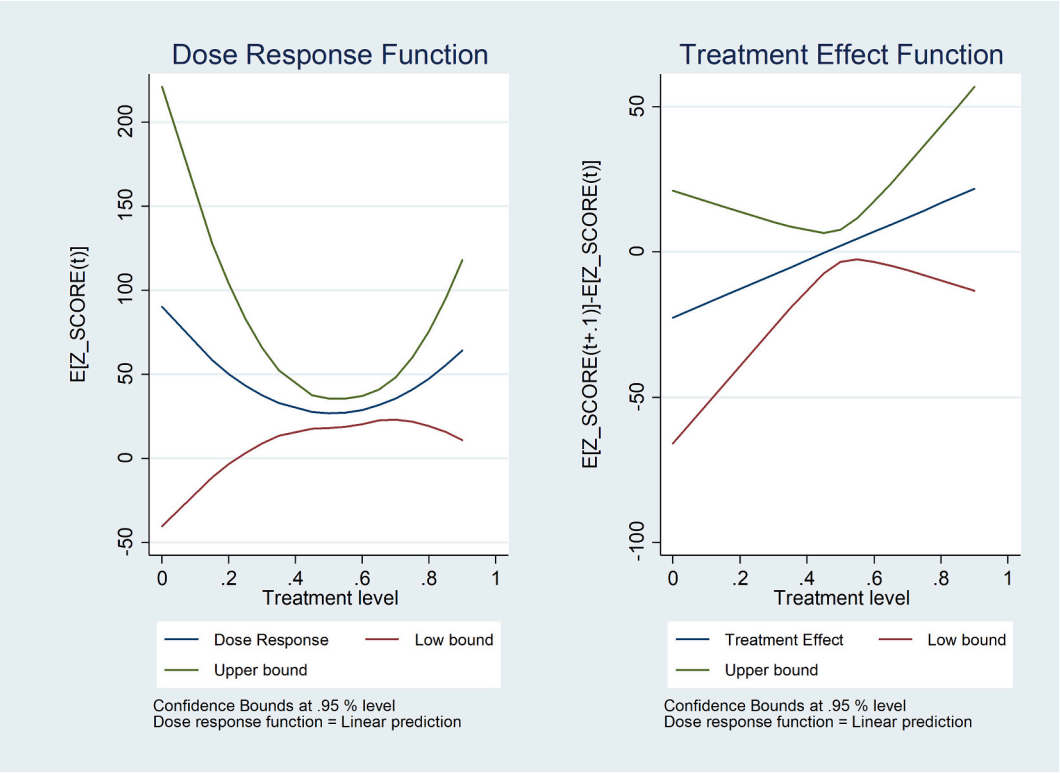


Fig. J.3. DRF and derivative DRF—z-score.

Appendix K. DRFs—Robustness check: Bank type (MENA sample)

.

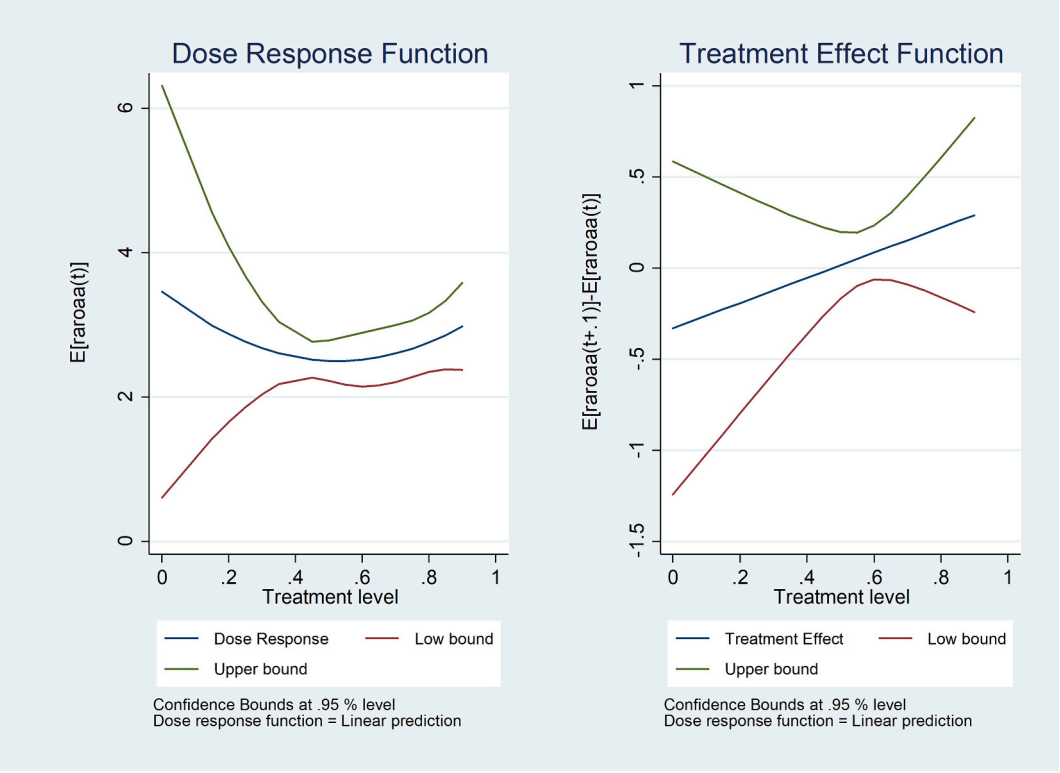


Fig. K.1. DRF and derivative DRF—raroaa.

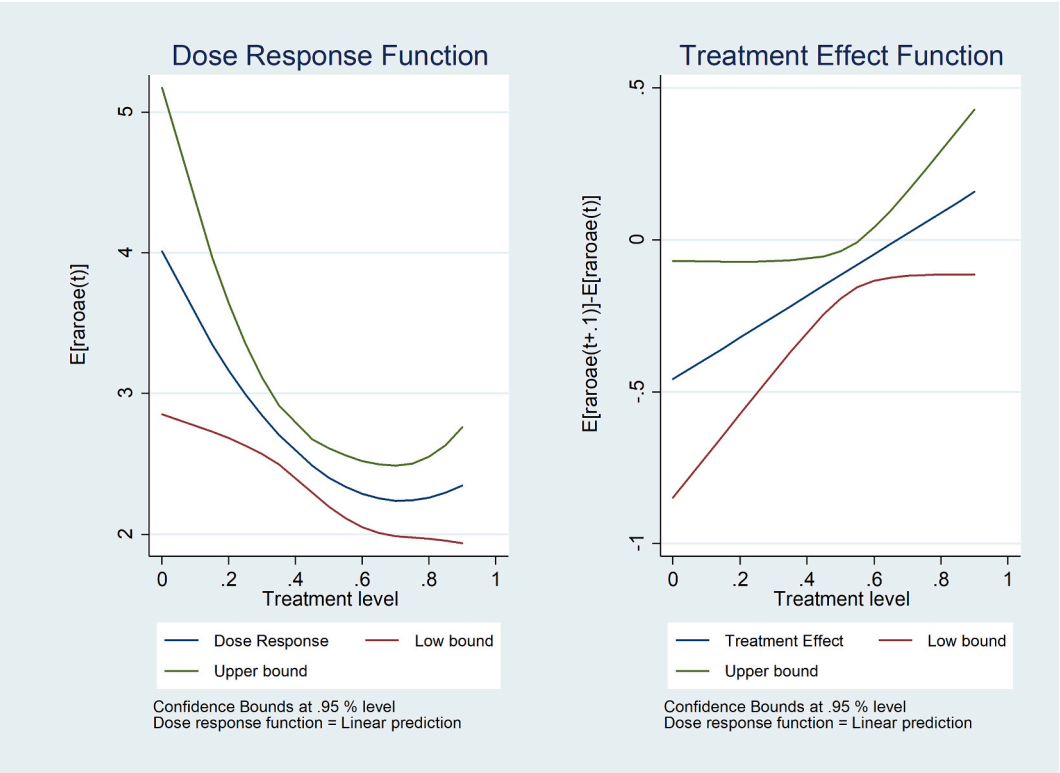


Fig. K.2. DRF and derivative DRF—raroae.

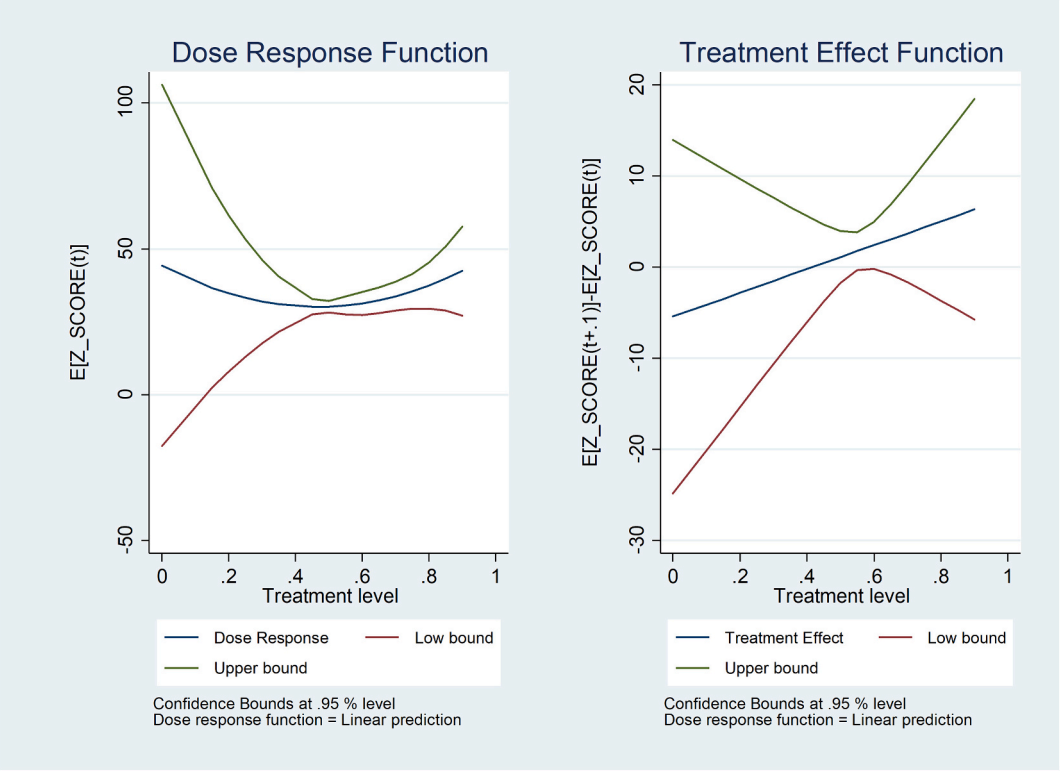


Fig. K.3. DRF and derivative DRF—z-score.

Appendix L. DRFs—Robustness check: Bank type (GCC sample)

.

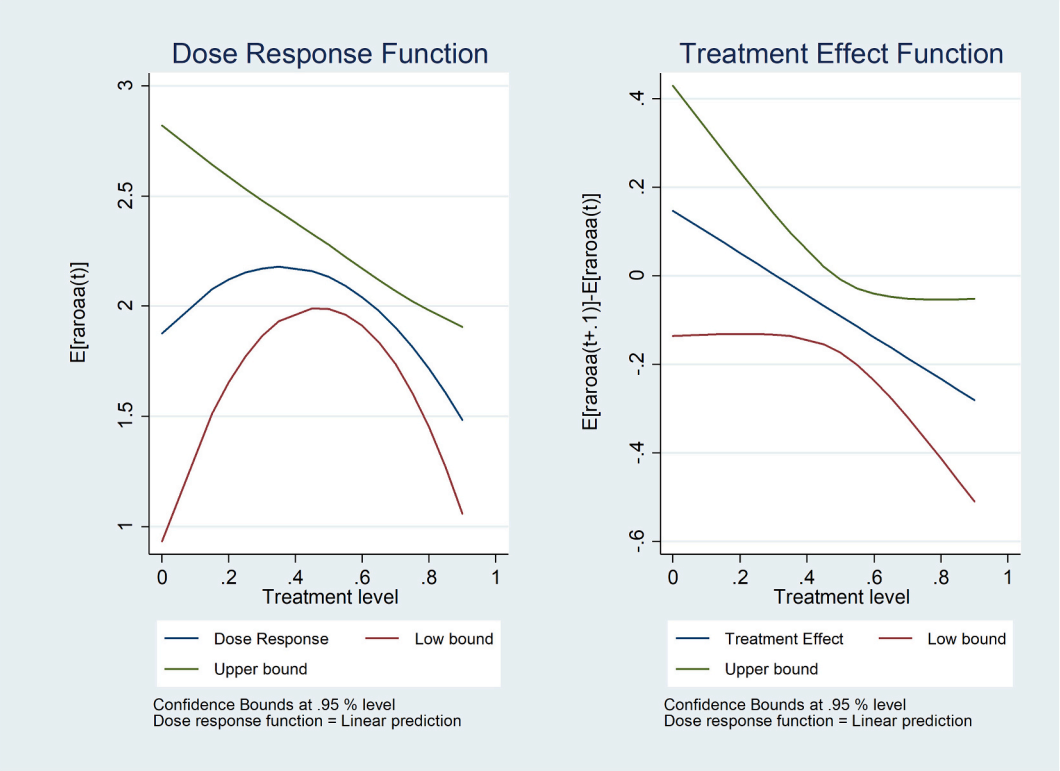


Fig. L.1. DRF and derivative DRF—raroaa.

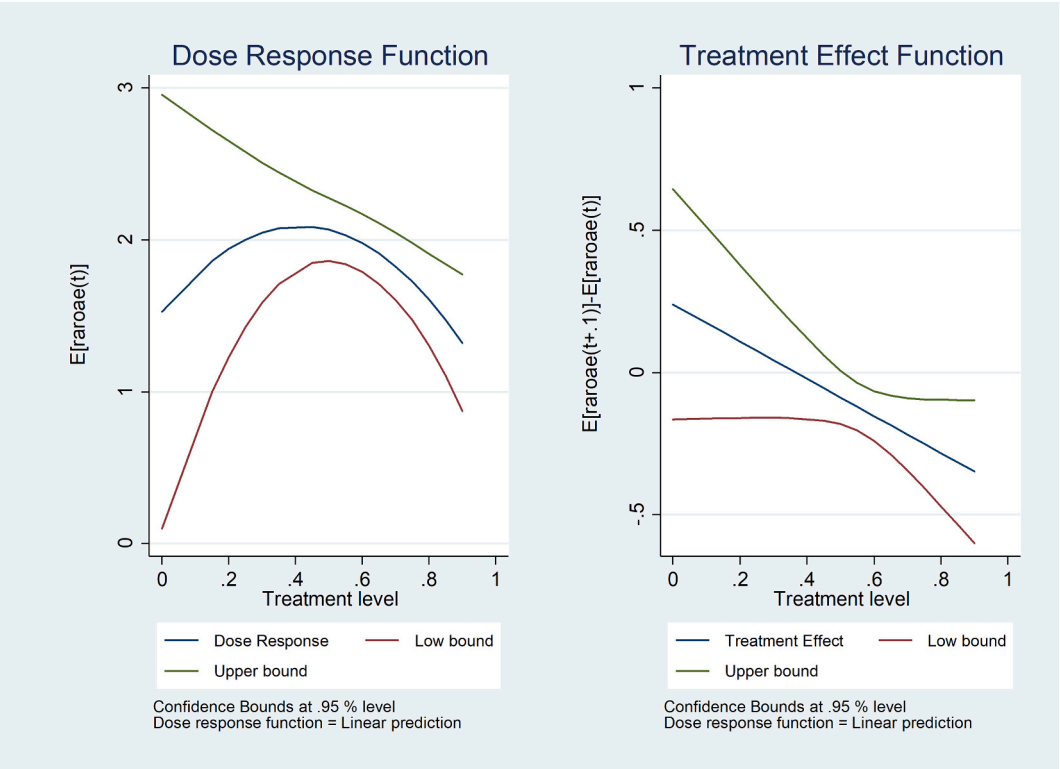


Fig. L.2. DRF and derivative DRF—raroae.

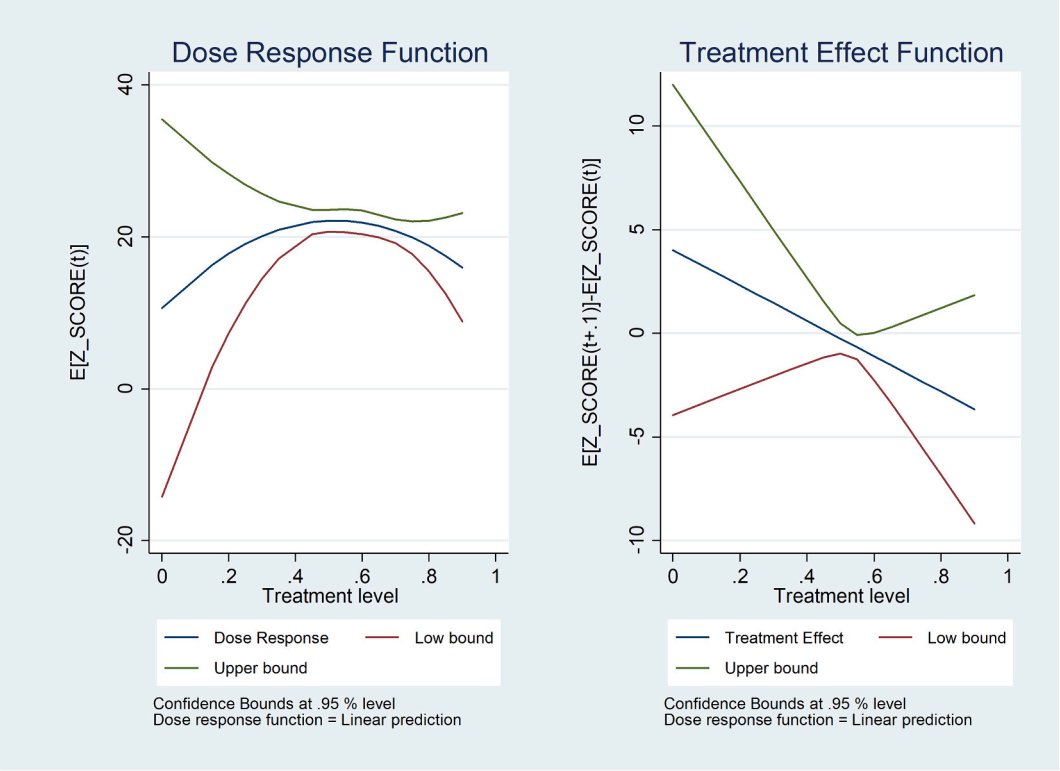


Fig. L.3. DRF and derivative DRF—z-score.

Appendix M. DRFs—Robustness check: Bank type (non-GCC sample)

.

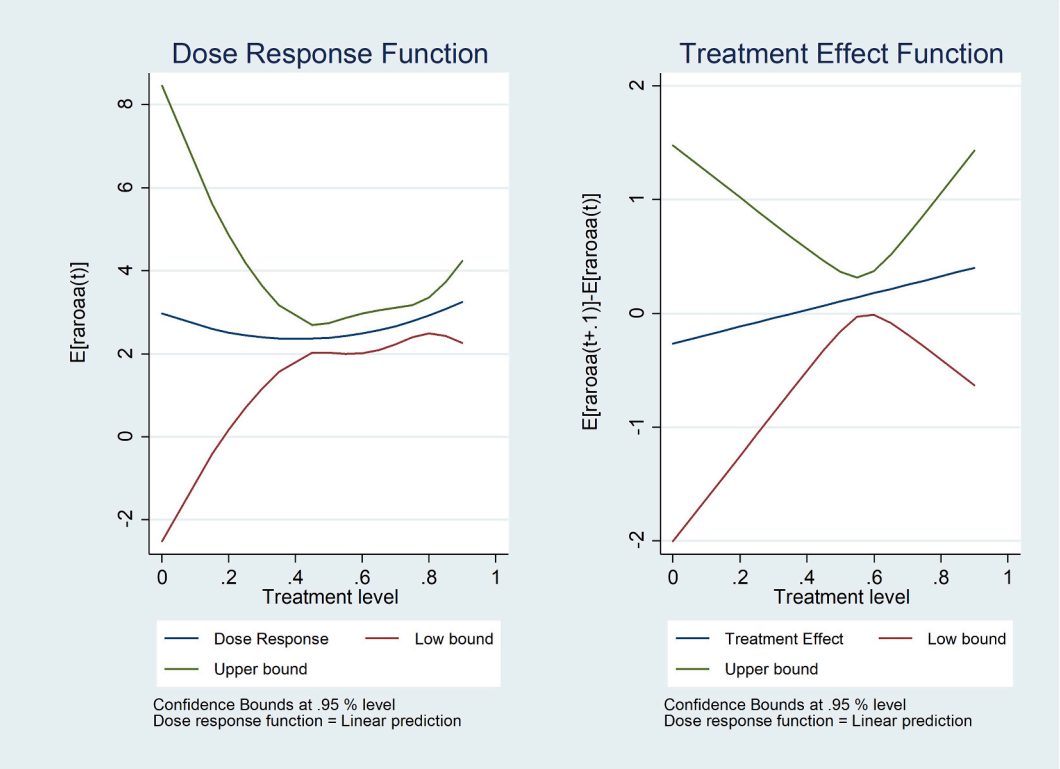


Fig. M.1. DRF and derivative DRF—raroaa.

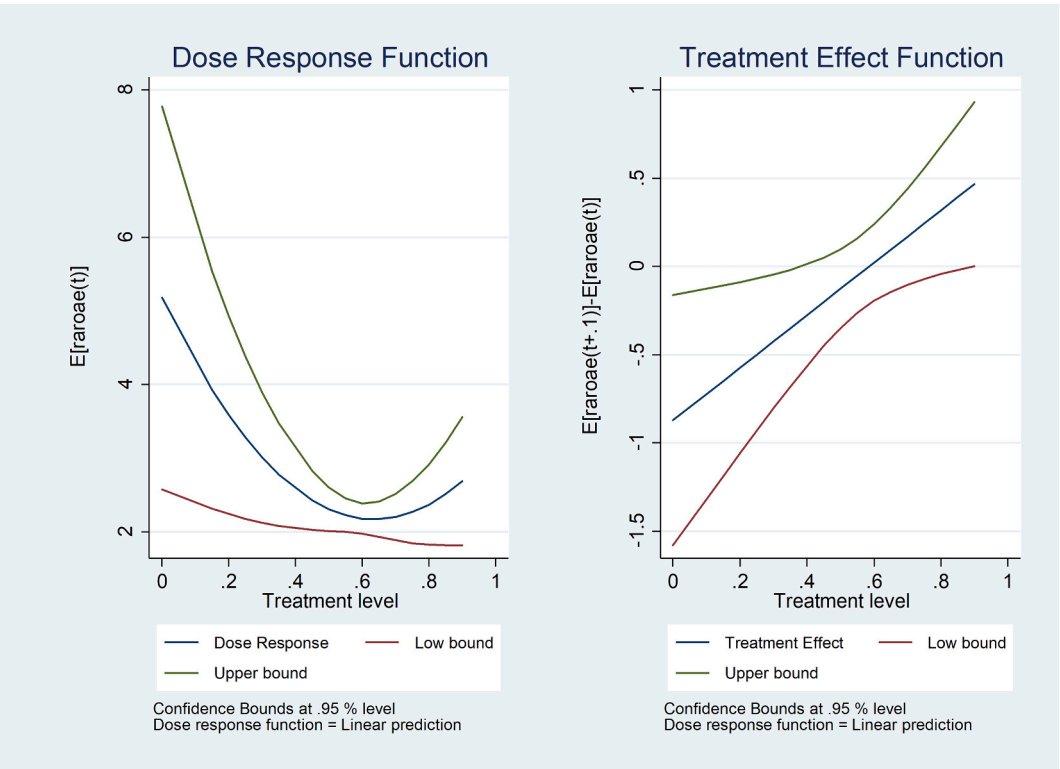


Fig. M.2. DRF and derivative DRF—raroae.

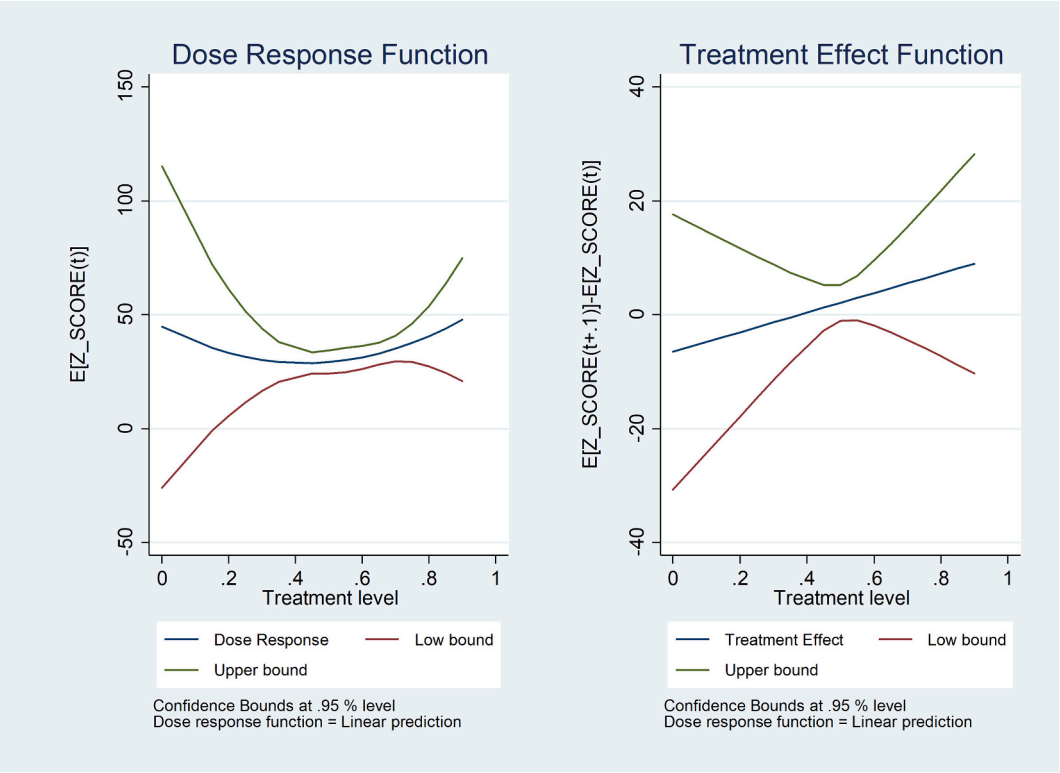


Fig. M.3. DRF and derivative DRF—z-score.

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